# Sumup - MS

## Coordinate Frames

### Inertial Frame (I-Frame)

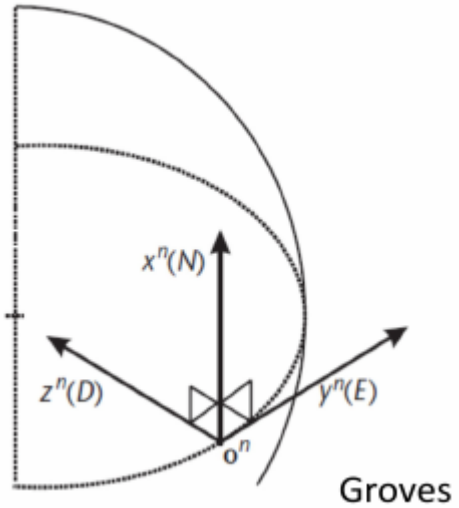
- **Inertial Frame**:
  - Frame without any acceleration or rotation with respect to the rest of the universe
- **Earth-Centered Inertial Frame (ECIF)**
  - Origin: Centre of mass of the Earth
  - z-axis: along the rotational axis through North
  - x,y-axes: in equatorial plane, fixed in space
- **Inertial sensors measure with respect to the inertial frame**

### Earth Centred Earth Fixed Frame (ECEF) (E-Frame)

- Similar to i-frame, but with x,y-axes also fixed to Earth
  - Origin: Center of mass = Earth center of mass
  - z-axis: along rotational axis ($z^e = z^i$) through North
  - x-axis: intersection of equatorial plane and reference meridian (Greenwich, zero latitude)
  - y-axis: completes right-handed system
- **E-frame rotates to the i-frame** along its **z-axis** with the **Earth Rotation Rate**
- **User position and velocity information** are usually given with respect to the **e-frame**

### Local Navigation Frame (N-Frame)

- Local system tangential to the ellipsoid of Earth model
  - origin: reference point of the user
    - e.g. centre of vehicle mass
  - z-axis: pointing down parallel to gravity
  - x-axis: towards North
  - y-axis: Completes right-handed system (East)
- Coordinates given as NED (North East Down)
  - Alternative defintions
- **N-frame** is usually used as a reference system for **attitude information**

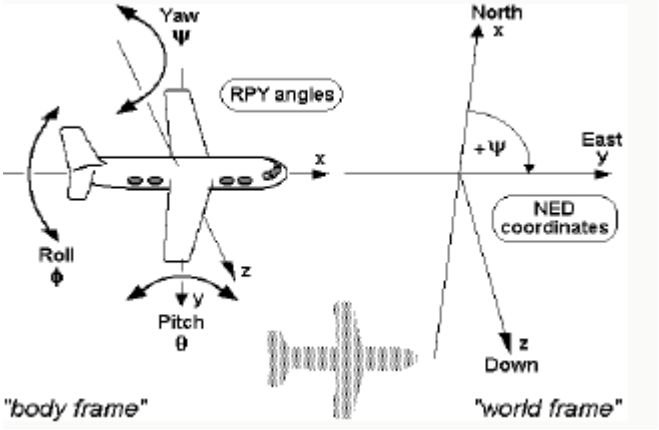


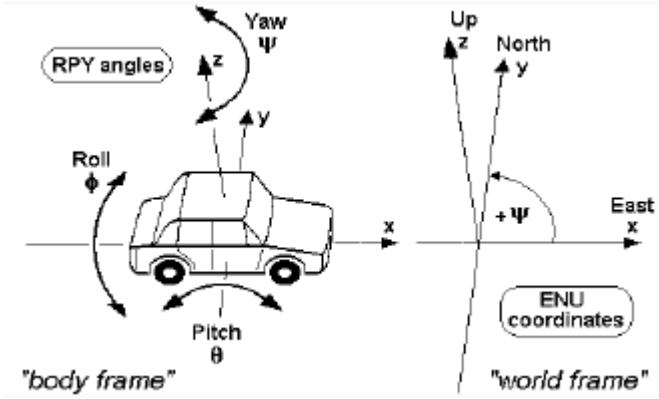

## Body Frame /Vehicle Frame (B-Frame)

- Attached to the user (e.g. vehicle)
  - origin: point on vehicle
    - e.g. centre of mass of vehicle
  - z-axis: pointing down
  - x-axis: pointing forward
  - y-axis: pointing to the right, completing the RHS
- Alternative definitions
- **Attitude** (Orientation) of a system is usually given as the **orientation** of the **b-fame relative to the n-frame**
- **Sensor data** are assumed to be available in the **b-frame**
  - (System Calibration!)

## Ground based/Aerial Vehicles

- Ground based and aerial vehicles usually have different definitions of the b- and n-frame
- **Aerial**:
  - n-frame: North-East-Down
  - b-frame:Forward-Right-Down
  - -> Heading towards: East: $\Psi = 90^\circ$

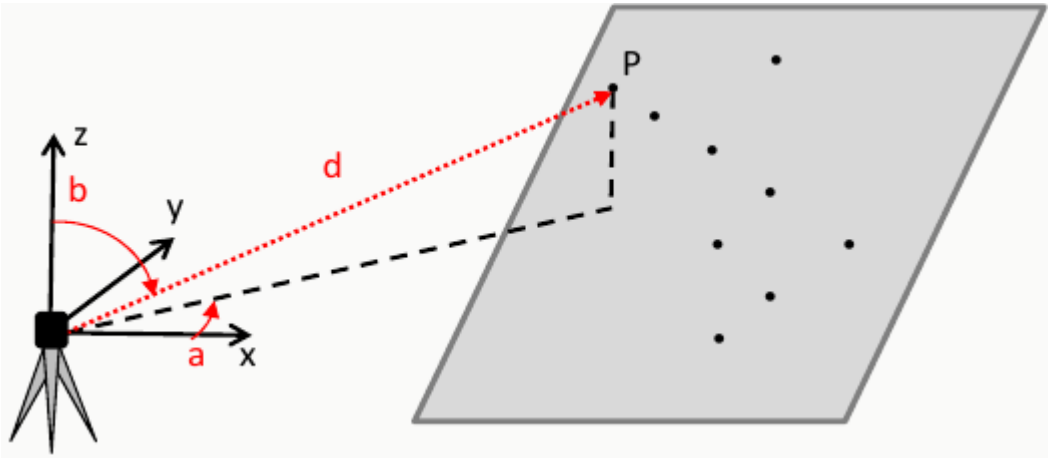
- 


- **Ground based**:
  - n-frame: East-North-Up
  - b-frame: Forward-left-up
  - Heading towards North: $\Psi = 90^\circ$
  - 


- All **angles zero** means **no rotation between b- and n-frame**
- *RHS-rule for both, but*
  - *direction of b-framefor aerial towards earth*
  - *North, East components changed for n-frame*

## Sensor Frame (S-Frame)

- S-Frame is the frame, in which sensor data is given
  - Origin and axis specific to the sensor
- Example: Panoramic Laser Scanner

- 


### Kinematic Quantities

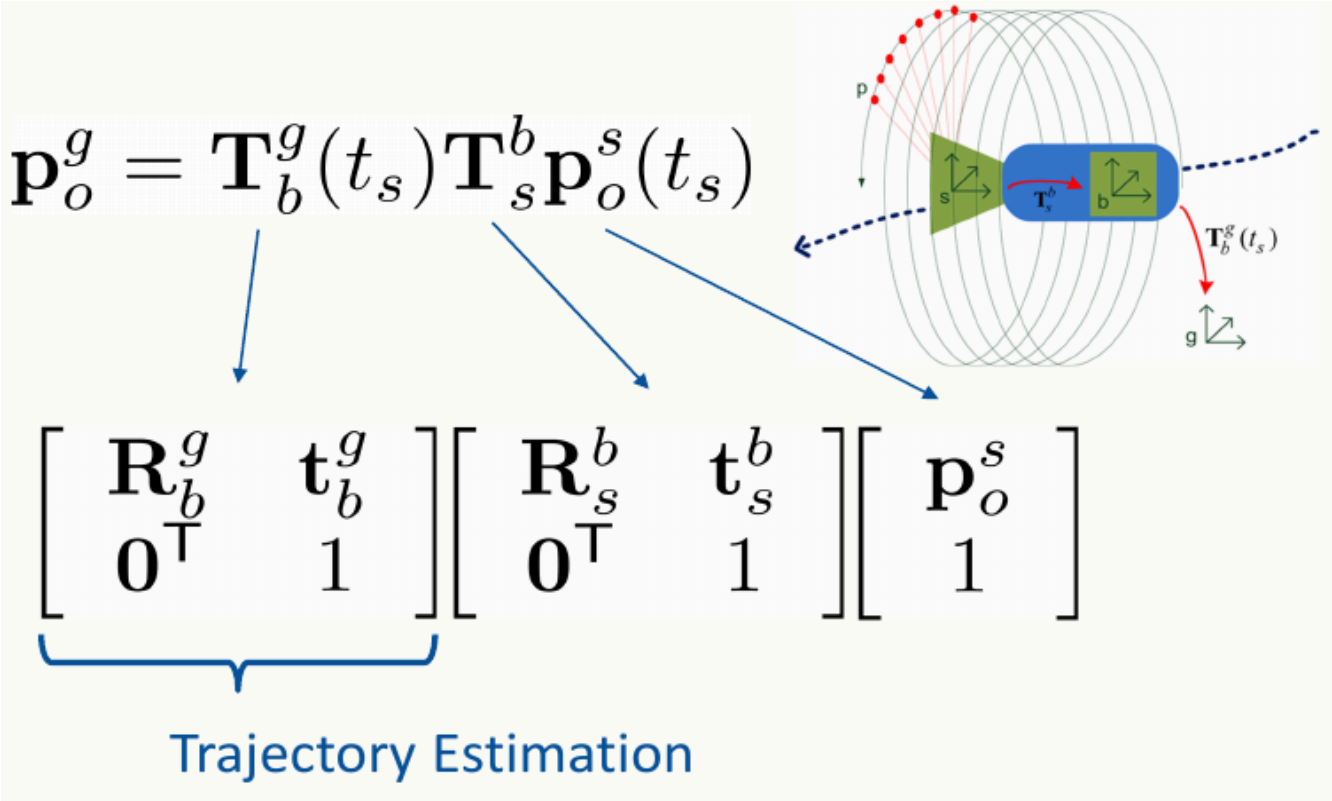
- Unambiguous definition of kinematic quantities such as position, velocity, acceleration and angular rate
  - velocity of the **b-frame**, relative to the **e-frame**, given in coordinates of the **n-frame**
    - $\boldsymbol{v}_{eb}^{n}$
  - Change of reference:
    - $\boldsymbol{v}_{eb}^{n} = -\boldsymbol{v}_{be}^{n}$
    - $\boldsymbol{\omega}_{eb}^{n} = -\boldsymbol{\omega}_{be}^{n}$
  - Concatenation:
    - $\boldsymbol{\omega}_{ib}^{b} = \boldsymbol{\omega}_{ie}^{b} + \boldsymbol{\omega}_{en}^{b} + \boldsymbol{\omega}_{nb}^{b}$
  - Change of resolving frame:
    - $\boldsymbol{\omega}_{nb}^{b} = \boldsymbol{C}_{n}^{b}\boldsymbol{\omega}_{nb}^{n}$

## Rotation Matrices

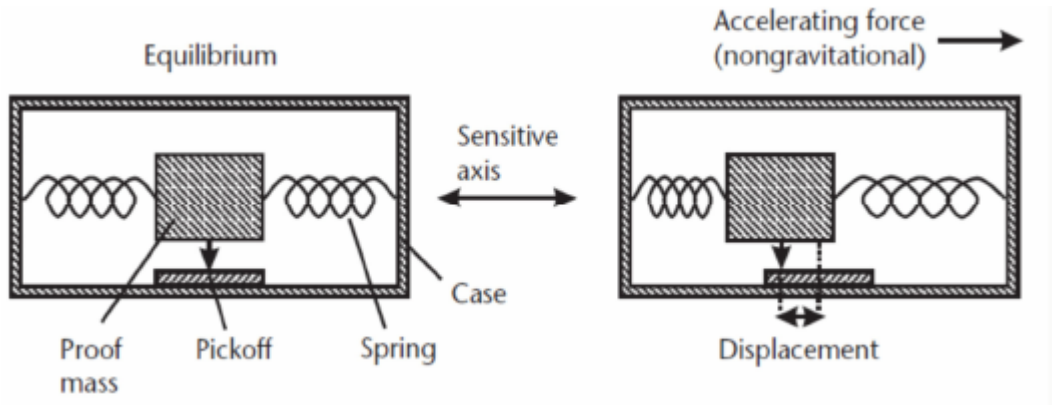
- Euler angles rotation rotate from n-frame to b-frame
  - $$\begin{aligned} \mathbf{C}_n^b &= \underset{\text{roll}}{\mathbf{R}_1(\phi)} \cdot \underset{\text{pitch}}{\mathbf{R}_2(\theta)} \cdot \underset{\text{yaw}}{\mathbf{R}_3(\psi)} \quad \text{3-2-1 convention} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

# Trajectory Estimation

- Georeferencing Measurement Data




## Inertial Sensors - Accelerometers

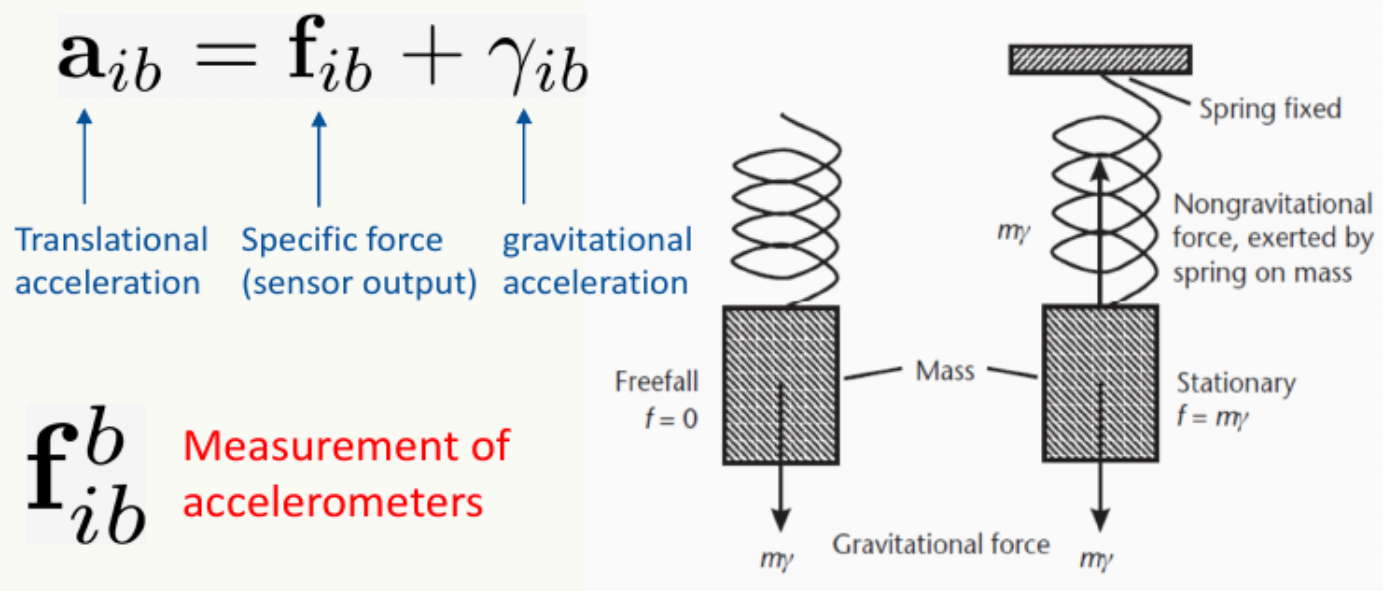
- A force $\boldsymbol{F}$ is applied to the sensor *box*
- Spring between box and proof mass $\boldsymbol{m}$
  - Applies force to mass, to compensate applied force
- 

- Hooks law:
  - $d \propto a$
- Displacement measurement via Capacity of Capacitors
  - Applied acceleration ->
  - Distance change ->
  - Capacity change ($C = \frac{A\epsilon_0}{d}$)->
  - Voltage change
- **Closed Loop** Pendulum Accelerometer

- Capacitative pickoff observes the displacement
- Necessary compensation current is **proportional** to applied acceleration

### Measurement of the Accelerometer

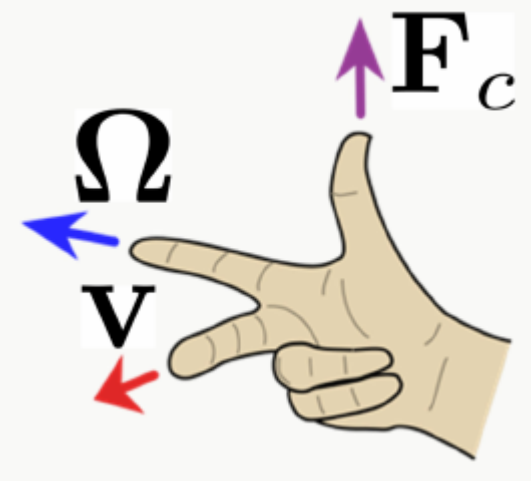
- Non-gravitational force per unit mass (~acceleration) with respect to inertial frame
  - (-> **Specific force** or **proper acceleration**)
    - Or **translational** or **coordinate acceleration**




## Gyroscope

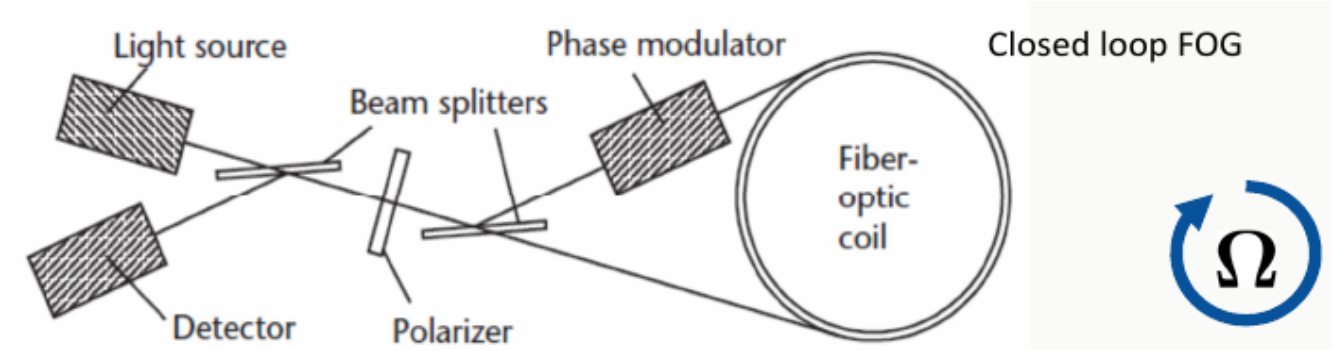
- The rotation axis of a spinning wheel gyroscope remains constant with respect to the inertial frame
- Rotation applied to the 2nd axis of the spinning wheel induce a rotation on the 3rd axis
- These gyroscope have been used to navigate accross the ocean without GPS and to fly to the moon and back

### Coriolis Force

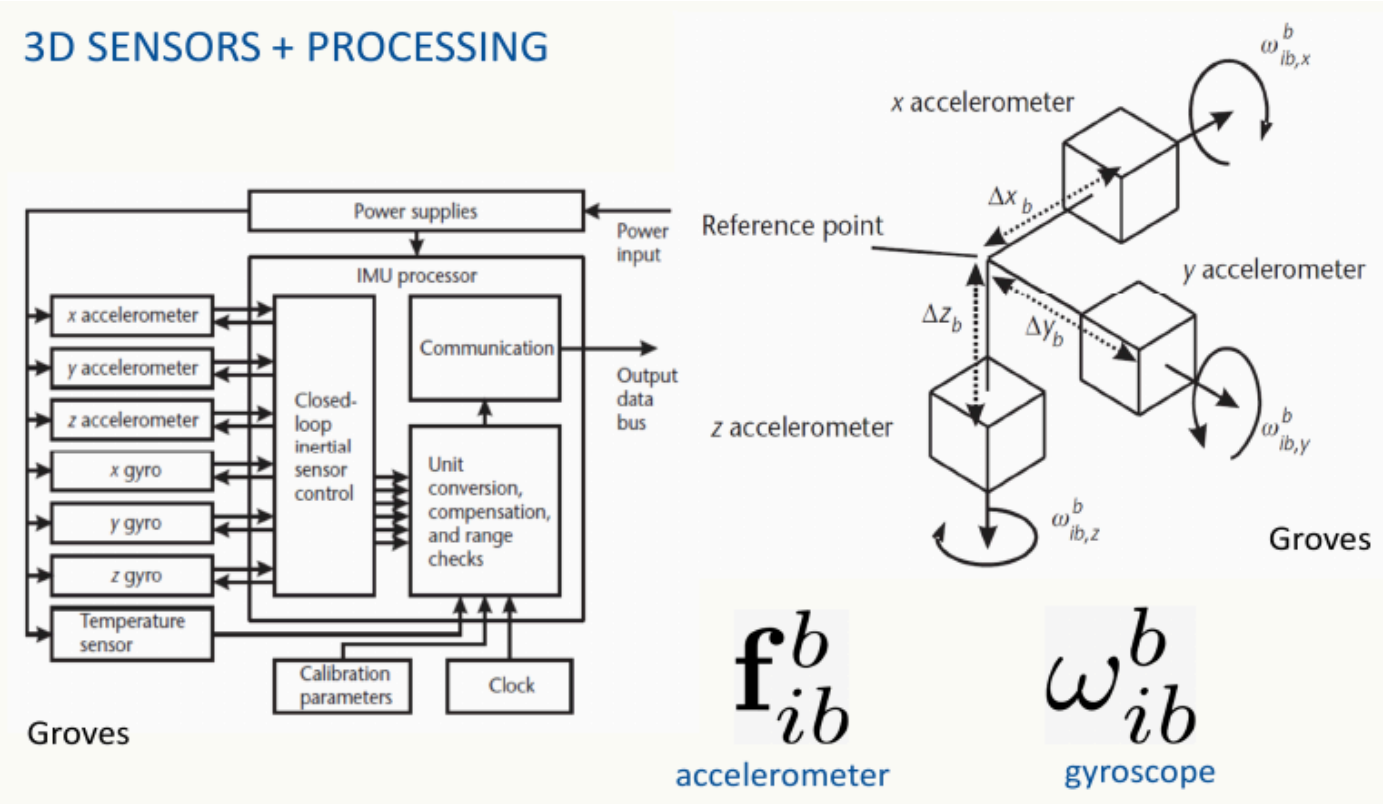



$$F_c = -2m(\Omega \times v)$$

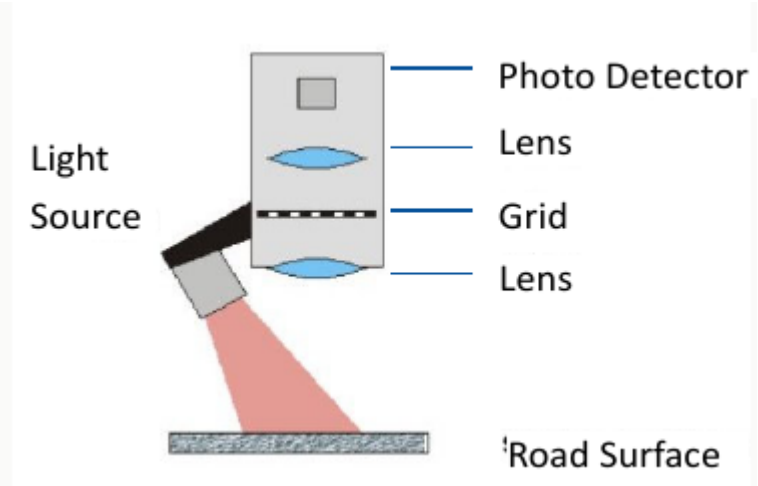
## Fiber-Optic Gyro (FOG)

- Light source is splitted and sent via both directions through a fiber-optic coil
- **Sagnac-effect** lead to phase change between the two beams
  - Ring/Coil has different RTT for different direction, if rotating
  - proportional to $\Omega$
  - Bigger area -> higher sensitivity
  - Coil of N rings multiplies the area by N
- Detector measures interference due to phase change
- **Closed loop**: Interference signal is kept constant by introducing an additional phase change in the modular
- 


## Inertial Measurement Units (IMU)



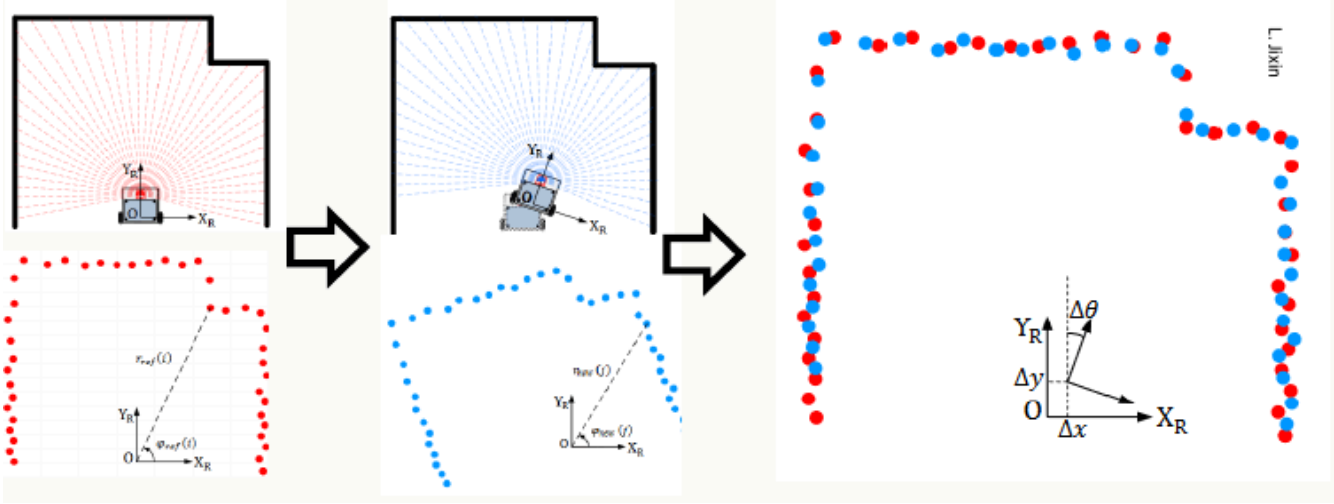

## Odometer

- Odometer: Measurement of the travelled distance
- Differential Odometer: Measurement travelled distance at 2 wheels
    - -> additional information about **direction**
- Measurement
    - Photo-electric transducer
    - Magnetband -> magnetic transducer
    - Cog wheel -> magnetic tranducers
- Correlation Based Velocity Measurement
    - 

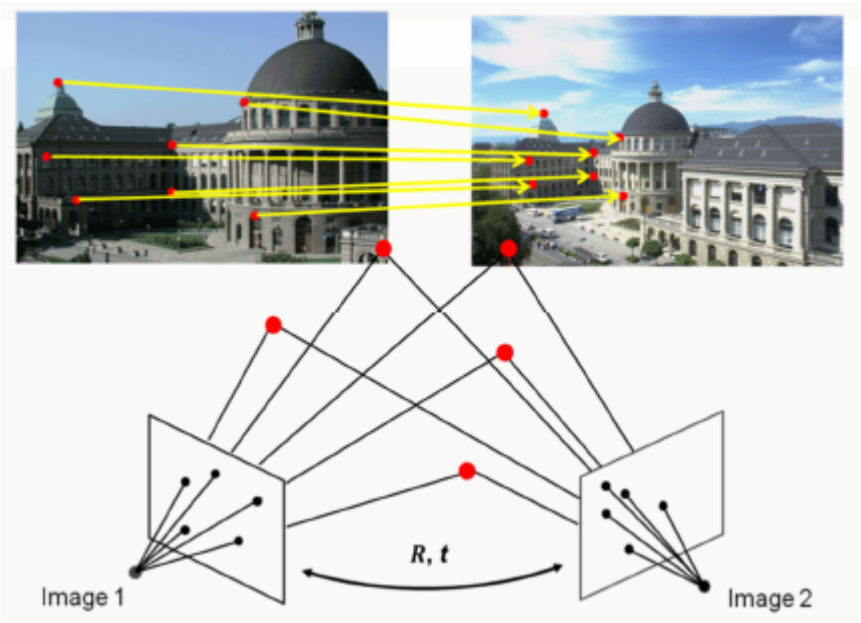
    - Cross correlation of 2 signals
        - time delay between the signals
        - velocity of the road surface relative to the sensors

## Scan Matching

- Laser Scanned Matching (-> Robotics)
- Assumptions:
    - Static Environment
    - Overlap between the scans
- Usual method:
    - Iterated Closest Point (ICP)

- 


## Visual Odometry

- Odometry Based on **Images**
  - relative motion between two cameras is calculated based on consecutive images
  - Features are calculated in one image
  - Calculate the features in the next images -> find correspondant features in previous image
  - Calculate transformation of the camera between the two images
  - -> relative information
  - -> Drift
- 

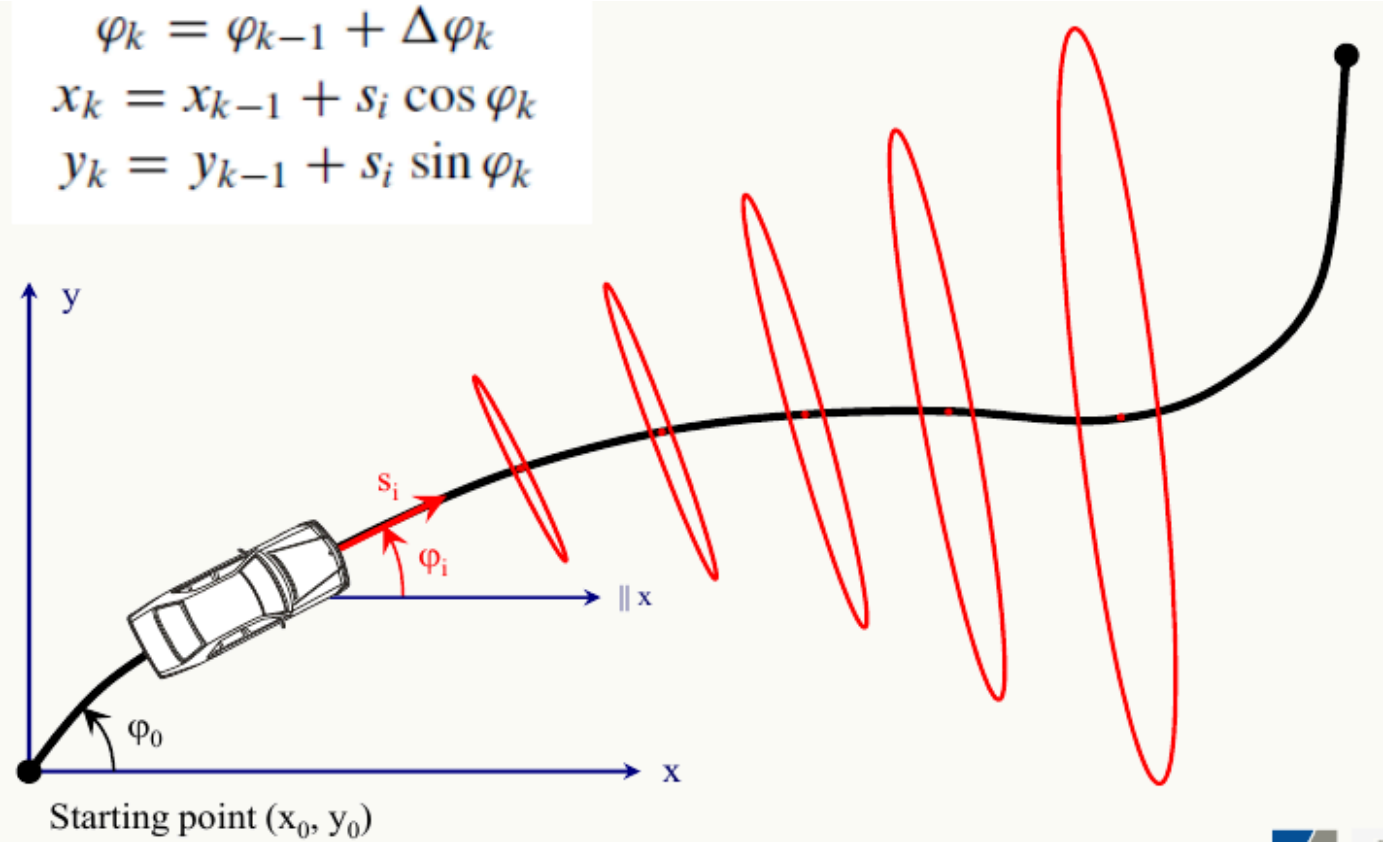

| Gyroscope | Accelerometer | Odometer | Differential odometer |
|---|---|---|---|
| Change in angle | Change in velocity | Travelled distance | Change in Direction |

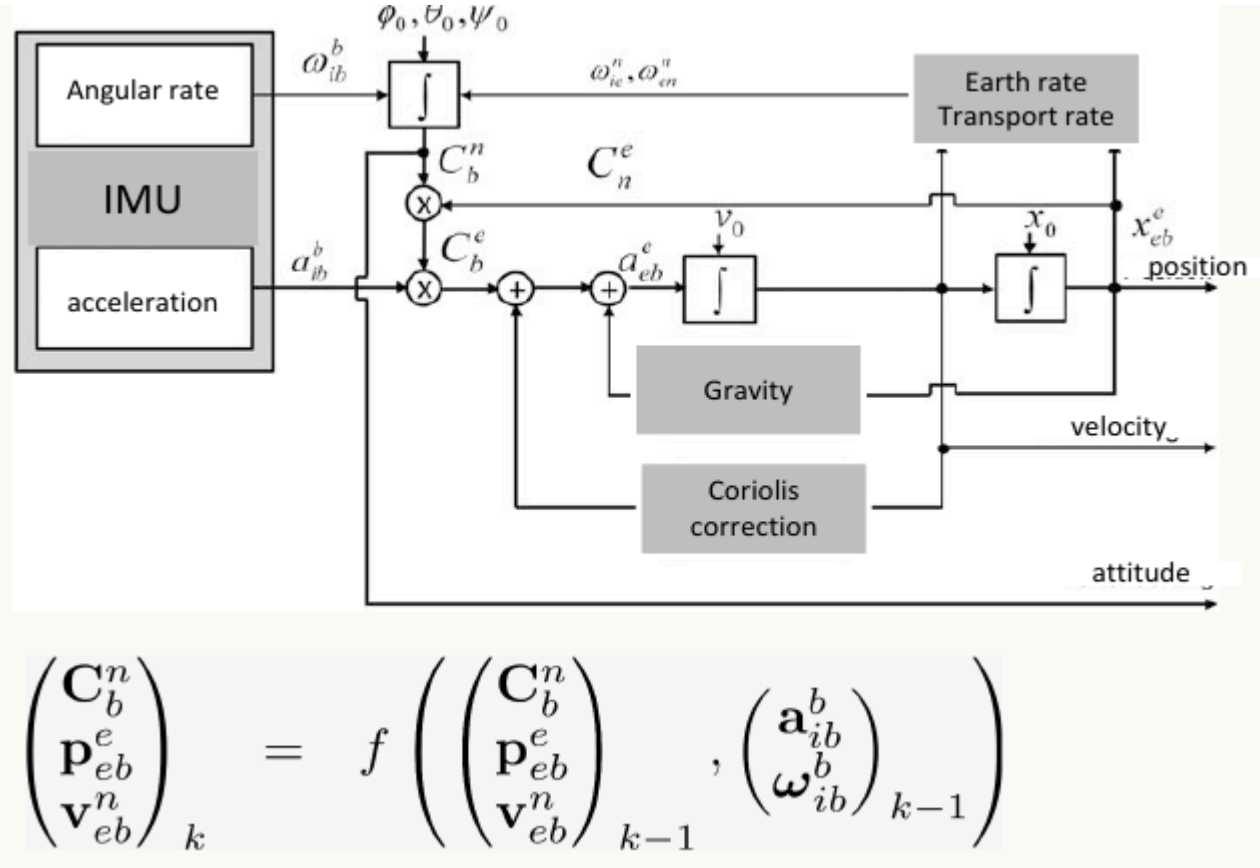
- 2D: Dead Reckoning
- 3D: Inertial Navigation

# Dead Reckoning

Position update using distance(Odometer) and direction (differential odometer) of travel

- **Direction**:
  - Compass
  - Angle to known landmarks
  - Angle to stars, celestial objects
  - Gyroscope (angular rate)
  - Differential odometer
  - **Angle** or integrated **angular rate** measurements
- **Speed**:
  - Counting steps (pedestrian navigation)
  - Counting knots on a rope (Ship)
  - Odometer
  - Accelerometer
  - **Distance travelled** per time or **velocity** over ground measurements






$$\begin{pmatrix} \mathbf{C}_b^n \\ \mathbf{p}_{eb}^e \\ \mathbf{v}_{eb}^n \end{pmatrix}_k = f\left( \begin{pmatrix} \mathbf{C}_b^n \\ \mathbf{p}_{eb}^e \\ \mathbf{v}_{eb}^n \end{pmatrix}_{k-1}, \begin{pmatrix} \mathbf{a}_{ib}^b \\ \boldsymbol{\omega}_{ib}^b \end{pmatrix}_{k-1} \right)$$

## 1D/2D Inertial Navigation

- e: earth fixed system
- b: body system
  - constraint to move on straight line pependicular to gravity -> no orientation
- accelerometer measure along the direction of motion, for now ignoring any motion of the Earth
- Numerical Integration
  - Midpoint or rectangular rule
  - Trapezoidal rule
  - More complicated and accurate integration schemes
    - e.g. Simpsons-Rule
- $v_{eb}(t) = v_{eb}(t_0) + \int_{t_0}^{t} a_{eb}(t')dt'$
- $r_{eb}(t) = r_{eb}(t_0) + \int_{t_0}^{t} v_{eb}(t')dt'$
- -> With **Numerical Integration**
- $v_{eb}(t_k) = v_{eb}(t_{k-1}) + \Delta t \cdot a_{eb}(t_k)dt$
- $r_{eb}(k) = r_{eb}(t_{k-1}) + \Delta t \cdot v_{eb}(t_k)dt$

### 2D Example

- Motion constraint to a plane perpendicular to gravity

- Orientation possible to any direction within the plan
- One angular and two linear degrees of freedom
- 2 accelerometers (sensitive axis horizontally along the body axis)
- acceleration $a^e_{eb,x}$ and $a^e_{eb,y}$ along the e-axis
  - **not** what the sensor measures:
  - sensor reading in b-frame
  - $a^b_{eb,x}$ and $a^b_{eb,y}$

**Inertial Navigation - 2D - No Gravity**

1. Attitude update $\Psi_{eb,k} = \Psi_{eb,k-1} + \Delta t \cdot \omega^b_{eb,k}$
2. Transformation of accelerometer measurement to the system of interest (**here e**)

$$\begin{pmatrix} a^e_{eb,x} \\ a^e_{eb,y} \end{pmatrix}_k = \begin{pmatrix} \cos\Psi_{eb,k} & -\sin\Psi_{eb,k} \\ \sin\Psi_{eb,k} & \cos\Psi_{eb,k} \end{pmatrix} \begin{pmatrix} a^b_{eb,x} \\ a^b_{eb,y} \end{pmatrix}_k$$

3. velocity update

$$\begin{pmatrix} v^e_{eb,x} \\ v^e_{eb,y} \end{pmatrix}_k = \begin{pmatrix} v^e_{eb,x} \\ v^e_{eb,y} \end{pmatrix}_{k-1} + \Delta t \cdot \begin{pmatrix} a^e_{eb,x} \\ a^e_{eb,y} \end{pmatrix}_k$$

4. Position update

$$\begin{pmatrix} r^e_{eb,x} \\ r^e_{eb,y} \end{pmatrix}_k = \begin{pmatrix} r^e_{eb,x} \\ r^e_{eb,y} \end{pmatrix}_{k-1} + \Delta t \cdot \begin{pmatrix} v^e_{eb,x} \\ v^e_{eb,y} \end{pmatrix}_k$$

Dead reckoning

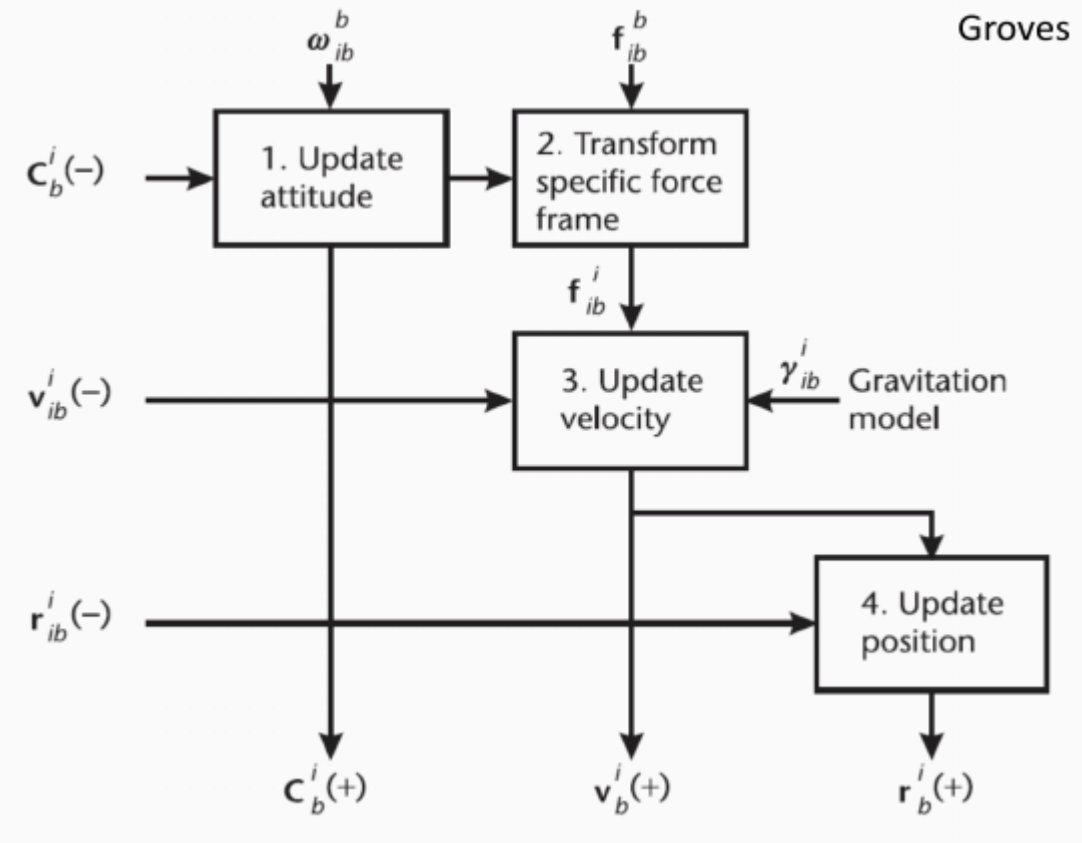
- From 2D to 3D:
  - Attitude update **needs integration of 3 angular** instead of 1
    - -> Integration of angular rates in 3D: NOT by integrating Euler angles independently
  - Gravitation not measured by accelerometer
    - Needs to be considered when modeling velocity
    - **Local gravity level** and **current orientation** with respect to gravity needs to be known
- Integration of Angular Rates in 3D

## 3D Inertial Navigation

- Wanted: position, velocity and attitude with respect to and resolved in the **inertial frame**
- Simple, because inertial sensors measure with respect to the inertial frame

- Only two frames are necessary, Earth rotation does not need to be considered

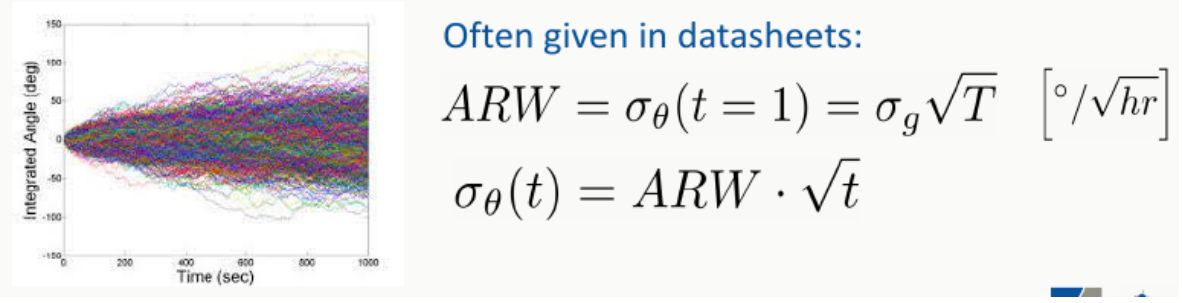
1. **Attitude update** (based on Gyroscope)
    - Integrate gyroscope readings to update the rotation matrix
    - Rotation matrix is defined between body and inertial frame (system of interest)
    - Gyroscopes measure angular rates between body and inertial frame
    - $\boldsymbol{C}_b^i(+) = \boldsymbol{C}_b^i(1_3 + \boldsymbol{\Omega}_{ib}^b \delta t)$
    - Approximated angular rates: $\boldsymbol{\Omega}_{ib}^b = \begin{bmatrix} 0 & -\omega_{ib,z}^b & \omega_{ib,y}^b \\ \omega_{ib,z}^b & 0 & \omega_{ib,x}^b \\ -\omega_{ib,y}^b & \omega_{ib,x}^b & 0 \end{bmatrix}$
2. **Transformation** of specific-force resolving axis (body frame) to the **coordinate frame** used for position and velocity (usually the e-frame)
    - Accelerometer measurements $\boldsymbol{f}_{ib}^b$ (specific force): b-frame -> inertial frame
    - Rotation matrix:
    - $\boldsymbol{f}_{ib}^i(t) = \boldsymbol{C}_b^i(+)\boldsymbol{f}_{ib}^b$
3. **Velocity Update**, including specific **force** AND **gravitational acceleration** (e.g. using a gravity model)
    - Assuming constant acceleration during update interval (integration of acceleration)
        - $\boldsymbol{v}_{ib}^i(+) = \boldsymbol{v}_{ib}^i(-) + \boldsymbol{a}_{ib}^i \delta t$
    - Acceleration consists of specific force (measured by sensor) And gravity
        - $\boldsymbol{a}_{ib}^i = \boldsymbol{f}_{ib}^i + \gamma_{ib}^i$
            - $\gamma_{ib}^i$ from gravity model
            - Or measured locally with known **rotation** and stationary sensor
4. **Position update**
    - Integrate acceleration
    - Assuming velocity is linear function of time during interval
    - $\boldsymbol{r}_{ib}^i(+) = r_{ib}^i(-) + v_{ib}^i(+)\delta t$




5. Optional. Transform attitude, velocity, position to useful frame
    - Use fancy super precise higher order integration schemes, if needed
    - Use implementation in other frames (Earth or local navigation frame), if needed
        - I to E-frame: Include Centrifugal Coriolis and Euler force in step 2 and additional rotation

# Error Propagation

## Ins Error Sources

- **Attitude Error**
    - $\theta_{err}(t) = \underbrace{b_{g0}t}_{\text{Start Bias}} + \underbrace{\int_0^t b_g'(\tau)d\tau}_{\text{Bias Instability}} + \underbrace{\int_0^t \epsilon_g(\tau)d\tau}_{\text{Angle Random Walk}}$
        - **White Noise sequence/random walk** with $T$ as Sampling Interval
        - 

        - Alternatively: Noise Density : $ND[(°/s)/\sqrt{Hz}]$
            - $\sigma_g = ND \cdot \sqrt{1/T} = ND/\sqrt{T} = ARW/\sqrt{T}$
            - PSD - Power Spectral Density: $PSD = ND^2$

- Bias has constant and changing part
  - Constant part can be different every time the sensor is switched on (**Turn-On Bias Stability**)
  - Changing Part: **Bias (In-)Stability** in datasheets in °/hr
    - Usually std over a time span of 100s
- Bias drift as random walk (not 100% true)

### Gyroscope Errors

$$\boldsymbol{\omega}_m = \mathbf{M}\mathbf{K}\boldsymbol{\omega}_t + \mathbf{b}_g + \epsilon_g$$

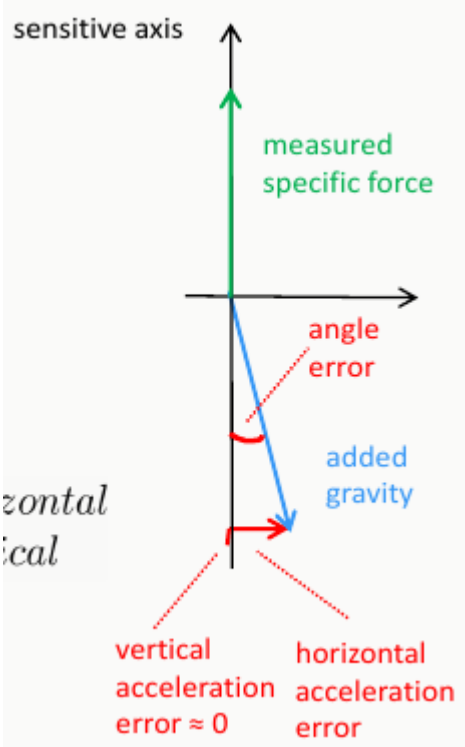
| RLG | FOG | MEMS |
|---|---|---|
| $0.001 - 10\,°/\mathrm{h}$ | $0.1 - 50\,°/\mathrm{h}$ | $5\,°/\mathrm{h} - 5\,°/\mathrm{s}$ |
| $5\,\mathrm{ppm}$ | $> 100\,\mathrm{ppm}$ | $> 400\,\mathrm{ppm}$ |
| $0.01\,°/\sqrt{\mathrm{h}}$ | $0.1\,°/\sqrt{\mathrm{h}}$ | $1\,°/\sqrt{\mathrm{h}}$ |

### Accelerometer errors

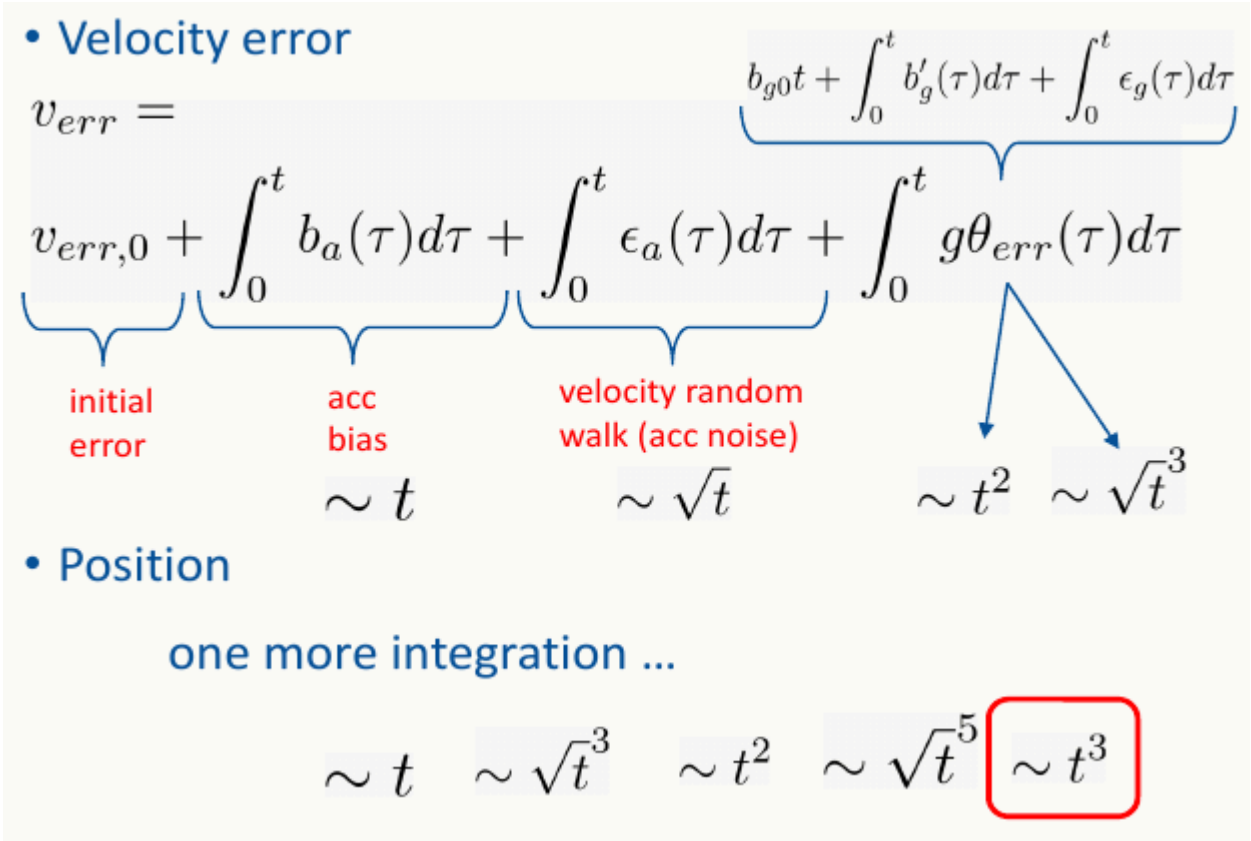
$$\mathbf{f}_m = \mathbf{M}\mathbf{K}\mathbf{f}_t + \mathbf{b}_a + \epsilon_a$$

| | Vibrating Beam | Pendel |
|---|---|---|
| constant | $0.1 - 1\,\mathrm{mg}$ | $0.1 - 10\,\mathrm{mg}$ |
| changing | $0.1\,\mathrm{mg}$ | $1\,\mathrm{mg}$ |
| | $100\,\mathrm{ppm}$ | $1000\,\mathrm{ppm}$ |
| Velocity RW | $0.01\,\mathrm{m/s}/\sqrt{h}$ | $0.04\,\mathrm{m/s}/\sqrt{h}$ |

- $\boldsymbol{a}_{ib}^n = \boldsymbol{C}_b^n \boldsymbol{f}_{ib}^b + \gamma^n$
  - accelerometer measurements
  - single axis
  - assume no scale factor error
  - angle error leads to vertical acceleration offset



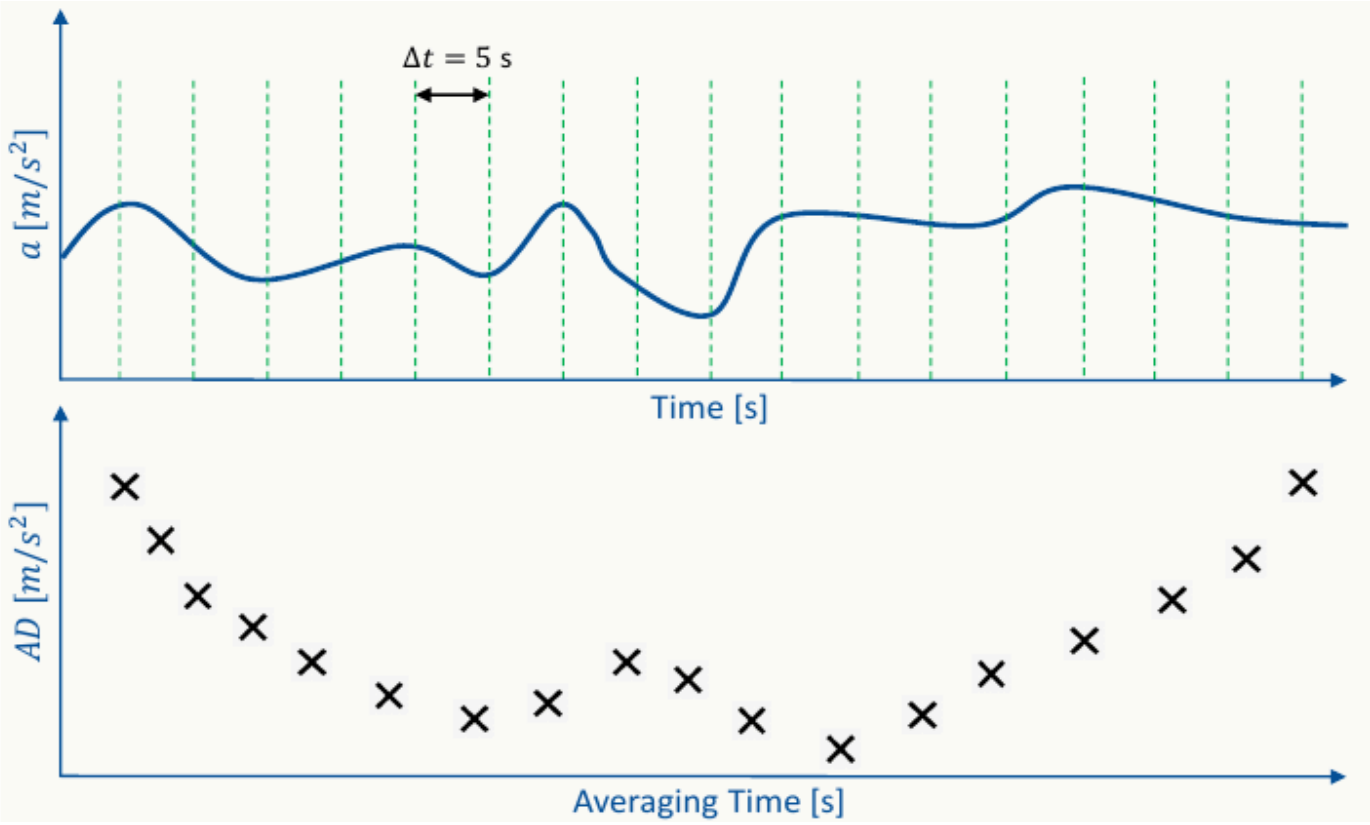

## Position and Velocity Error



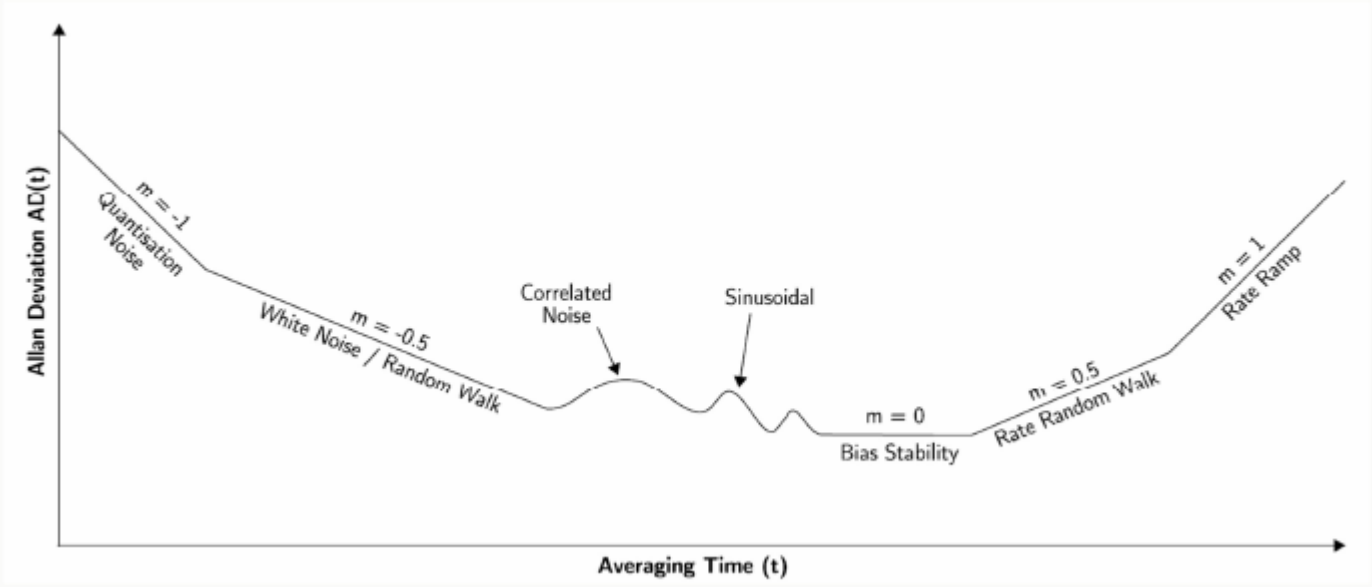

## Allan Variance

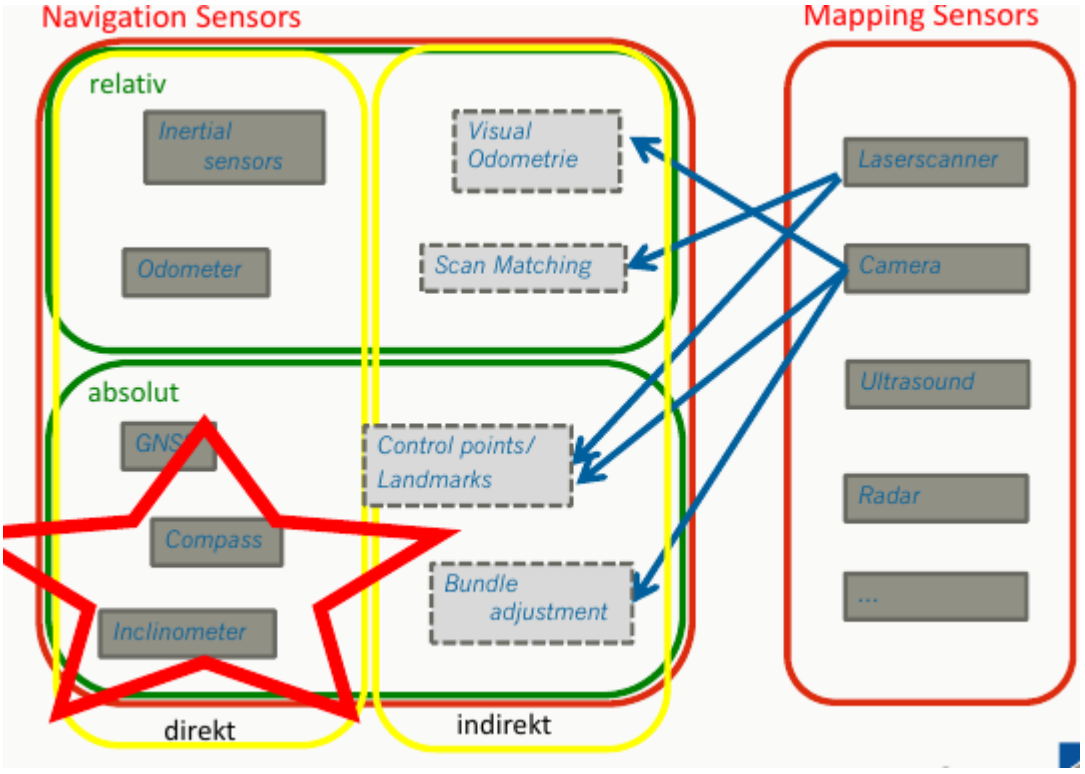
- Useful to quantify noise processes in inertial sensors
- Calculation
  - Build intervals of different length
    - $[a(\Delta t)_1, a(\Delta t)_2, ...a(\Delta t)_n]$

- Calculate variance of consecutive interval mean differences
  - $\sigma_{al}^2(\Delta t) = \frac{1}{2\cdot(n-1)} \sum_i (a(\Delta t)_{t+1} - a(\Delta t)_i)^2$



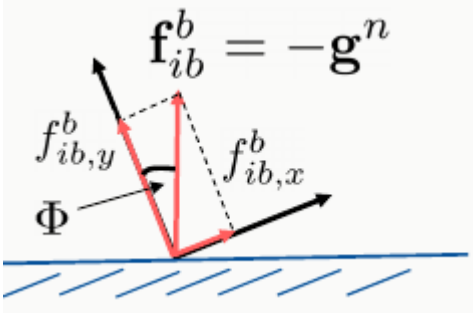

## Allan Deviation (Idealised)

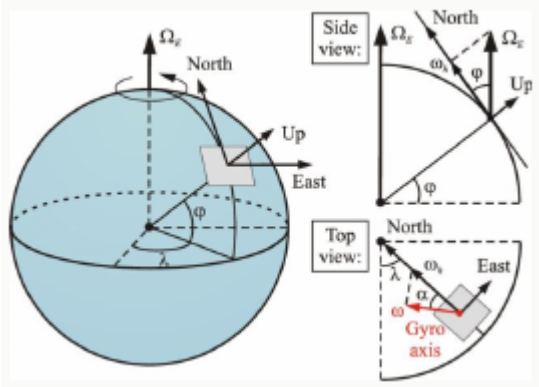





## Initialisation & Alignment

- **Position**:
  - Use absolute sensors (e.g. GNSS)
  - Start from a known position
- **Velocity**:
  - Use absolute sensor (e.g. GNSS)
  - Start in a static situation (v=0)
- **Orientation**:
  - Magnetic Compass
  - Inclinometer or accelerometer
  - Gyroscopic compass
  - Start from a known orientation
  - Orientation from vector observations
    - Observe a vector $\boldsymbol{v}^e$, which is constant with respect to the Earth in the local coordinate system of the sensor $\boldsymbol{v}^b$
    - Find rotation matrix $\boldsymbol{C}_b^e$ with $\boldsymbol{v}^e = \boldsymbol{C}_b^e \boldsymbol{v}^b$
    - Not solvable with one vector observation (one angle remains underdetermined)
  - -> get only 2 angles OR use multiple vectors
- Options: gravity, magnetic field, Earth rotation rate
- **Accelerometer Levelling**

- **In stationary systems**, roll and pitch can be derived from accelerometer
  - $\phi_{nb} = arctan(\frac{f^b_{ib,y}}{f^b_{ib,x}})$
  - 


- **Gyroscope Compassing**
  - Use Earth rotation rate to measure yaw angle
  - Not possible with low cost IMUs, because Earth rate is below noise
  - 

  - Roll and Pitch from accelerometer leveling
- **Magnetic Compassing**
  - Use compass to measure the 3D Earth magnetic field vector $\boldsymbol{m}^b_{mb}$
  - Roll and Pitch from accelerometer leveling ($\boldsymbol{\Theta}_{nb}, \boldsymbol{\Phi}_{nb}$)
  - Yaw is measured with respect to magnetic north! ($\Psi_{nb} = \Psi_{mb} + \alpha_{nE}$)

## GNSS in Mobile Multi Sensor Systems

- Changing conditions
  - No averaging
  - Jumps
  - **Multipath**
  - **Non-line-of-sight signals**
- Use **Carrier phase based relative positioning**
  - ambiguity fixing difficult for inaccurate float solutions
  - ambiguities have to be fixed again after every loss
- Use of **reference services**

## Orientation

- Using a sequence of position
  - Orientation vs. rotation
  - Track $\neq$ Heading
  - Orientation $\tilde{=}$ needs 2 epoches
  - -> Multi Antenna systems are better !
- Multi-Antenna System
  - Two Baselinevectors
    - Yaw- and Pitch-angle
    - Roll-angle
  - Accuracy depends on baseline length

## GPS Velocity Determination

- Possible observations:
  - RTK-Position of a single antenna (differentiation)
  - Doppler observations generated by the receivers
  - **Doppler measurement derived from carrier phase observations**
- $\dot{\phi}_k^j = \frac{\phi_{k+\Delta t}^j - \phi_{k-\Delta t}^j}{s\Delta t}$
- No ambiguity resolution necessary
- Results in 3D-velocity vector, if position and sat position are used
- Accuracy for 1-10Hz measurement rate
  - 1 mm/s - 2 cm/s

# Kalman Filter

The temporal evolution of the state is described by the **system model** (sometimes called **process model**)

$$\mathbf{x}_k = \mathbf{\Phi}_k \mathbf{x}_{k-1} + \mathbf{B}_k \mathbf{u}_k + \mathbf{w}_k$$

$\mathbf{\Phi}_k$ The **state transition matrix** describes the effect of the old state on the new state

$\mathbf{u}_k$ **control input** vector (e.g. throttle, break, steering)

$\mathbf{B}_k$ The **control input matrix** describes the effect of some control input on the state

$\mathbf{w}_k$ The **process noise** represents all uncertainties of the system model and is assumed to be normally distributed with covariance $\mathbf{Q}_k$

1D train

$$\mathbf{x}_k = \begin{bmatrix} r_k \\ \dot{r}_k \end{bmatrix} \quad \mathbf{u}_k = \frac{f_k}{m}$$

$f_k$: external applied force (e.g. brake or throttle)

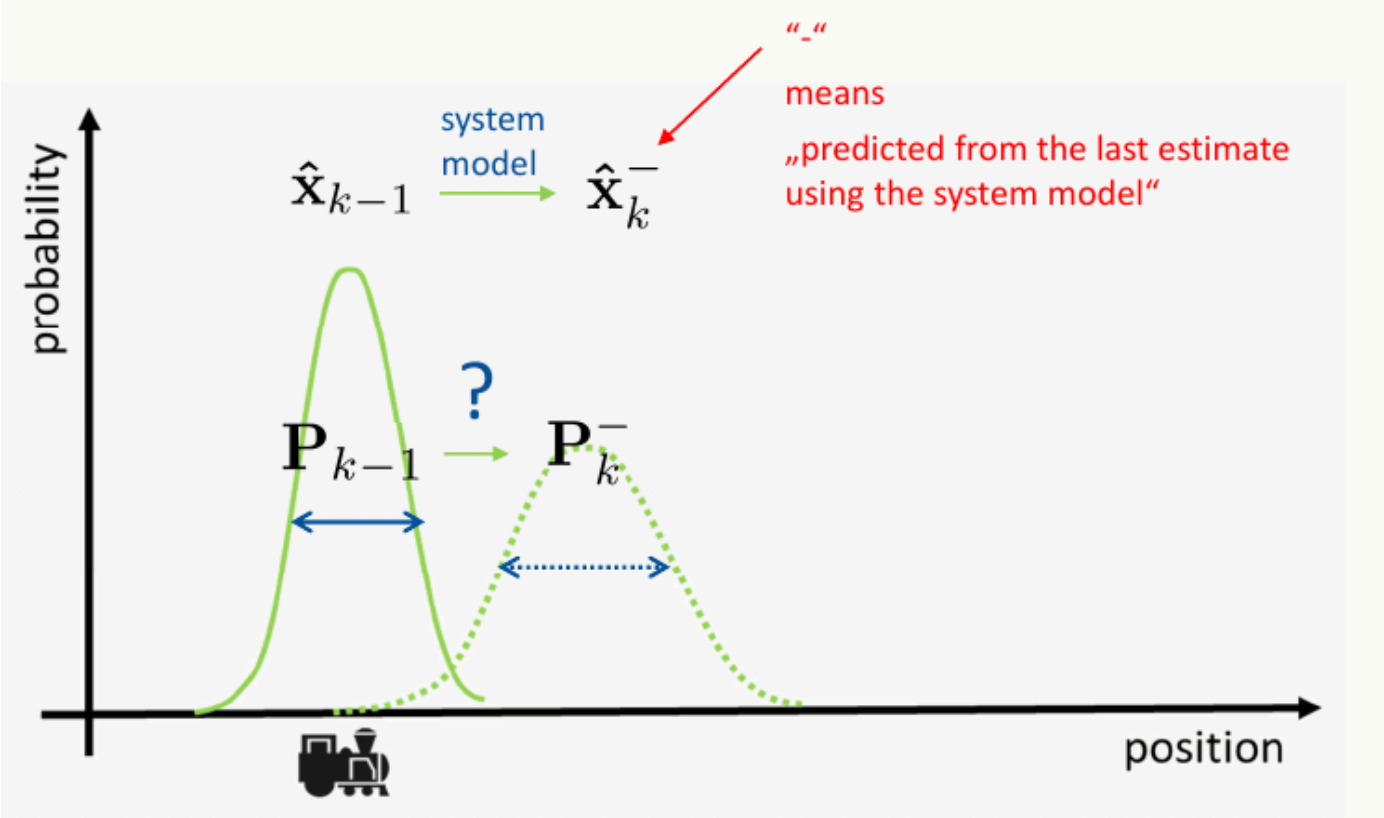
$m$: mass of the system

From Newtons equations of motion:

$$r_k = r_{k-1} + \dot{r}_{k-1} \cdot \Delta t + \frac{1}{2}\frac{f_k}{m}(\Delta t)^2$$
$$\dot{r}_k = \dot{r}_{k-1} + \frac{f_k}{m}\Delta t$$

Can also be derived from differential equation systems
➔ see literature

$$\underbrace{\begin{bmatrix} r_k \\ \dot{r}_k \end{bmatrix}}_{\mathbf{x}_k} = \underbrace{\begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}}_{\mathbf{\Phi}_k} \underbrace{\begin{bmatrix} r_{k-1} \\ \dot{r}_{k-1} \end{bmatrix}}_{\mathbf{x}_{k-1}} + \underbrace{\begin{bmatrix} \frac{1}{2}(\Delta t)^2 \\ \Delta t \end{bmatrix}}_{\mathbf{B}_k} \underbrace{\frac{f_k}{m}}_{\mathbf{u}_k}$$




- After predicting $\hat{\boldsymbol{x}}_{\boldsymbol{k}}^-$:
- $\boldsymbol{P}_k^- = \boldsymbol{\Phi}_k \boldsymbol{P}_{k-1} \boldsymbol{\Phi}_k^T + \boldsymbol{Q}_k$

- A way to describe the **measurement** (including uncertainty)
  - The **measurement model** (sometimes also called **observation model**) is describing what the measurement for a given state vector is

$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k$$

$\mathbf{z}_k$ The **measurement vector** contains the measurements taken at a time $t_k$

$\mathbf{H}_k$ **The measurement** or **observation matrix** is linking the state to the measurements

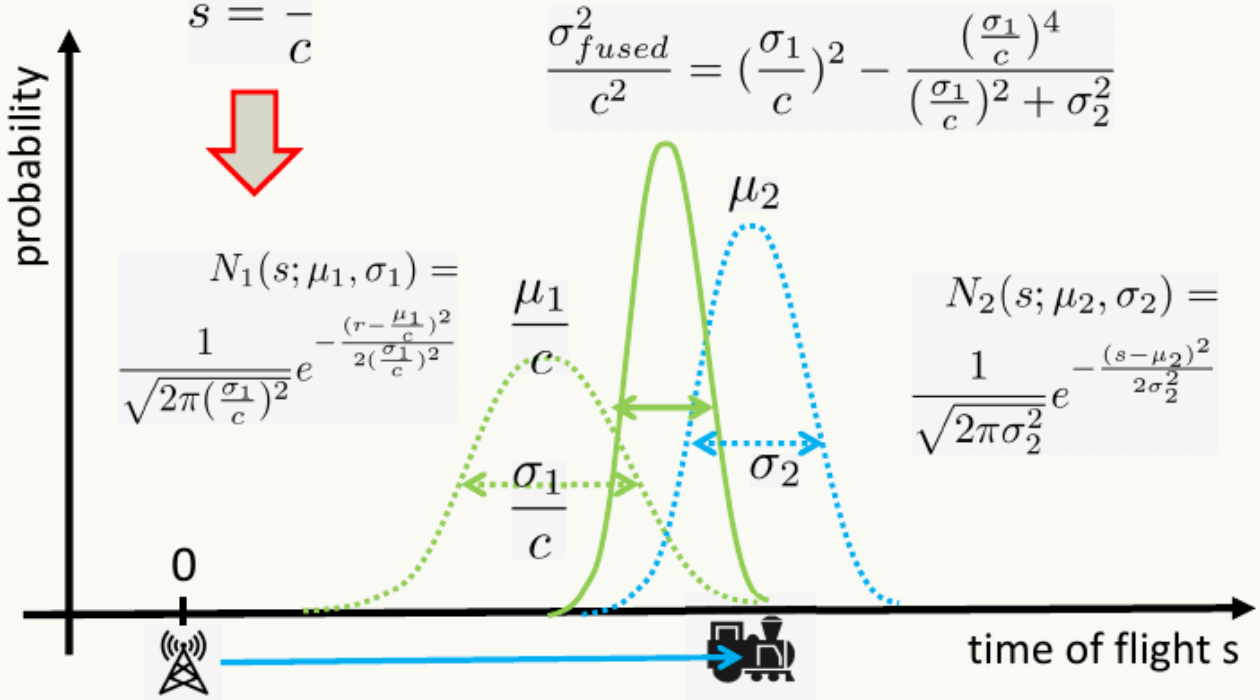
$\mathbf{v}_k$ The **measurment noise** represents the uncertainties of the measurement model and is assumed to be normally distributed with covariance $\mathbf{R}_k$

c = speed of ligth

$$\frac{\mu_{fused}}{c} = \frac{\mu_1}{c} + \frac{(\frac{\sigma_1}{c})^2}{(\frac{\sigma_1}{c})^2 + \sigma_2^2}(\mu_2 - \frac{\mu_1}{c})$$

$$\frac{\sigma_{fused}^2}{c^2} = (\frac{\sigma_1}{c})^2 - \frac{(\frac{\sigma_1}{c})^4}{(\frac{\sigma_1}{c})^2 + \sigma_2^2}$$

$$s = \frac{r}{c}$$




measurement model $\mathbf{H}_k$

predicted covariance $\mathbf{P}_k^-$

$$K = \frac{H\sigma_1^2}{H^2\sigma_1^2 + \sigma_2^2}$$

$\mathbf{K}_k$ Kalman gain

$\mathbf{R}_k$ measurement noise

predicted state $\hat{\mathbf{x}}_k^-$

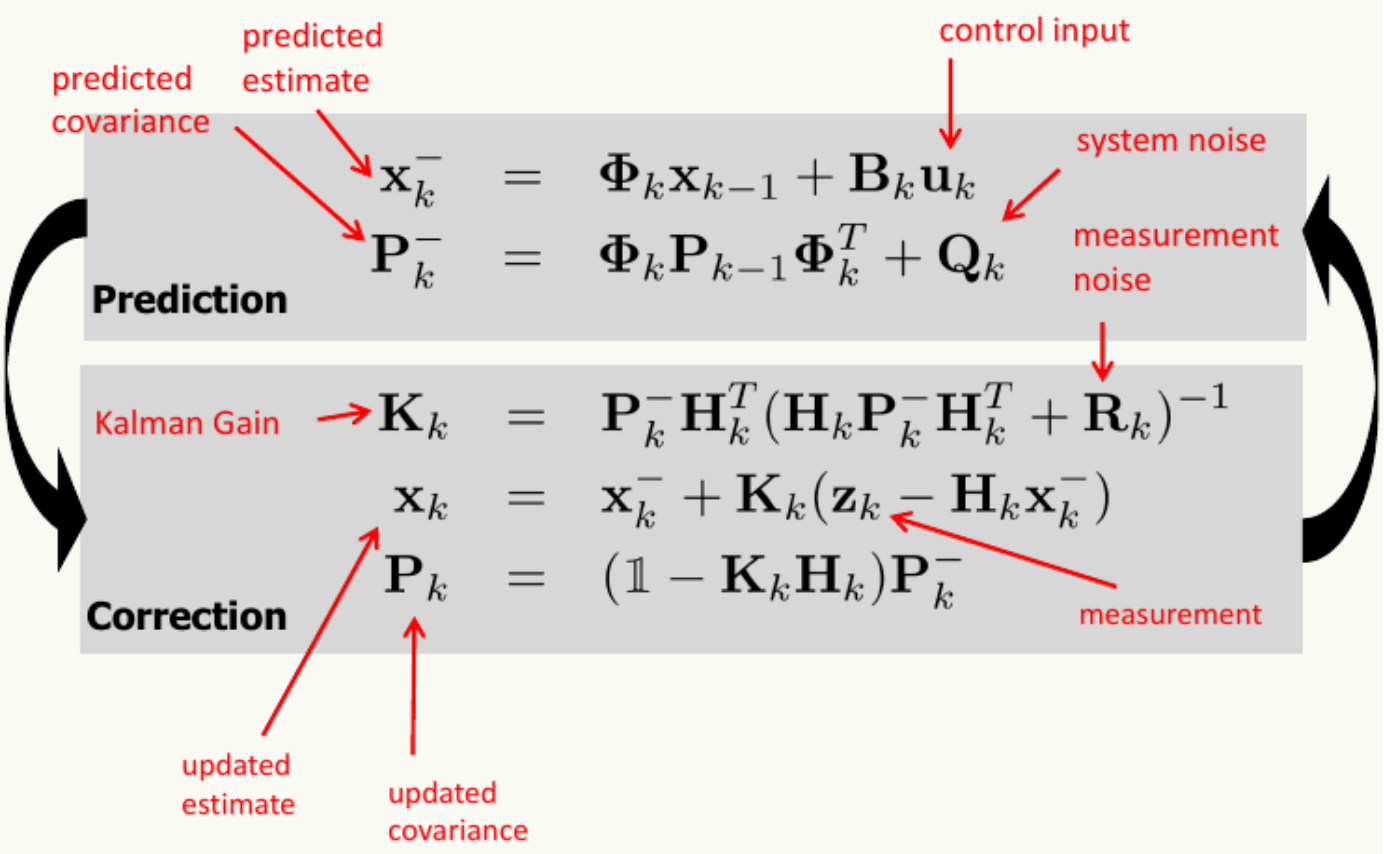
$\mathbf{z}_k$ measurement

$\hat{\mathbf{x}}_k$ updated state

$$\mu_{fused} = \mu_1 + K(\mu_2 - H\mu_1)$$

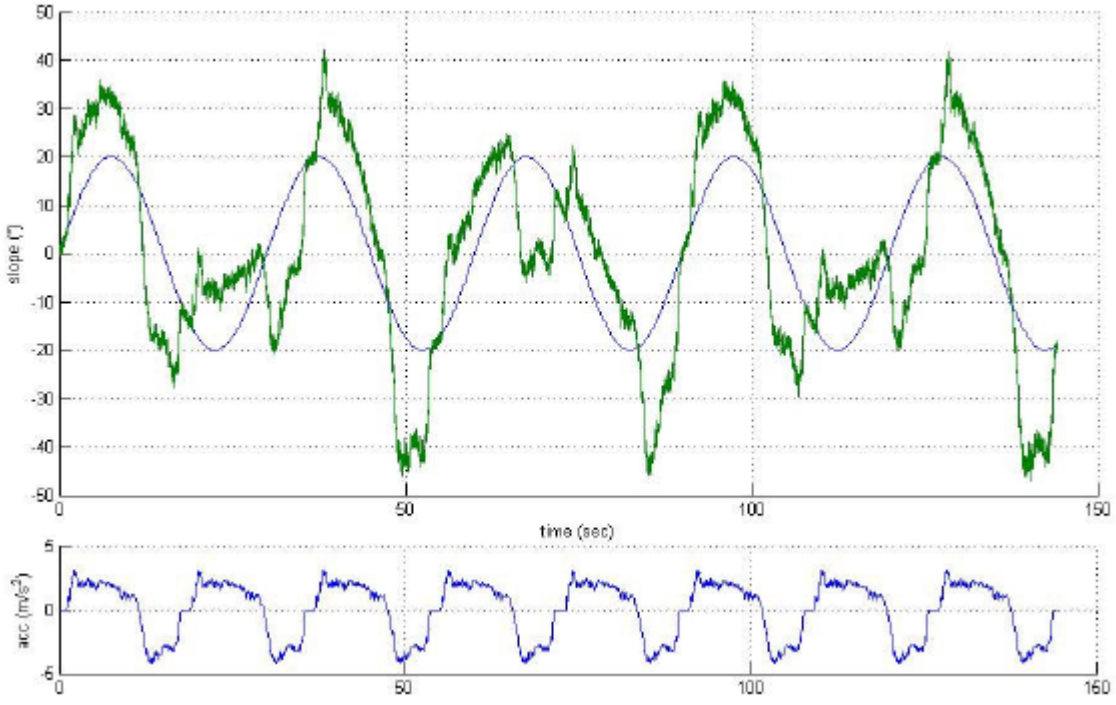
$$\sigma_{fused}^2 = \sigma_1^2 - KH\sigma_1^2$$

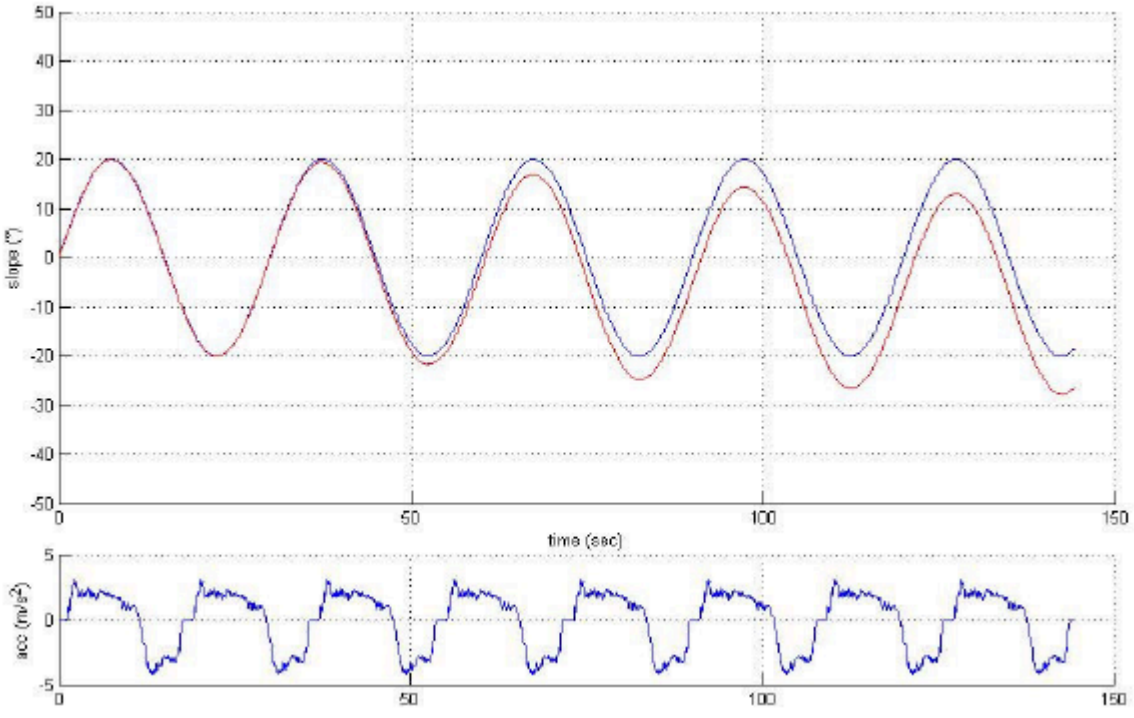
updated covariance $\mathbf{P}_k$

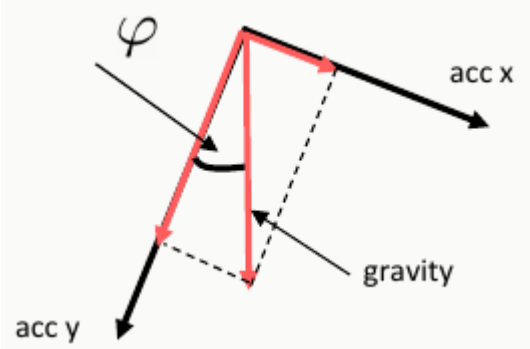



## Relative Sensor Readings as Control Input

- **Accelerometer**:
  - $\varphi_{acc} = atan(\frac{acc_x}{acc_y})$
  - **Problem**: Sensitive to additional acceleration due to motion
  - Accelerometer only:
    - Long term stable measurements (drift free)
    - strong short disturbances due to acceleration/deceleration

- 


- **Gyroscope**:
  - $\varphi_{gyro} = \varphi_0 + \int_{t_0}^{t} w(t)dt$
  - **Problem**:
    - Errors accumulating (drift)
    - Unknown Initial Value
    - Gyroscope only (Strapdown Integration)
      - Accurate short term measurements, but drift over time
    - 

- 

  - System model:

$$
\begin{array}{ccccccc}
\boldsymbol{x}_k & = & \boldsymbol{\Phi}_k & \boldsymbol{x}_{k-1} & + & \boldsymbol{B}_k & \boldsymbol{u}_k + \boldsymbol{w}_k \\
\downarrow & & \downarrow & \downarrow & & \downarrow & \downarrow \\
\underset{\text{angle}}{\varphi_k} & = & 1 & \cdot\varphi_{k-1} & + & \underset{\text{sampling time}}{\tau_s} \cdot & \underset{\text{angular rate}}{w_k}
\end{array}
$$

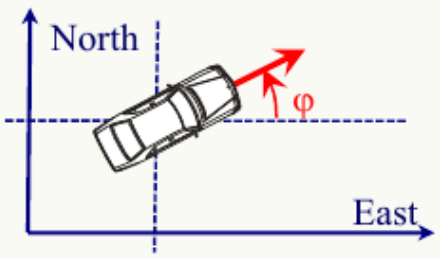
- Measurement Model

$$
\begin{array}{c}
\boldsymbol{z}_k = \boldsymbol{H}_k \boldsymbol{x}_k + \boldsymbol{v}_k \\
\downarrow \quad \downarrow \\
\varphi_{acc} = 1 \cdot \varphi_k = atan(\frac{acc_x}{acc_y})
\end{array}
$$

## Kalman Filter to fuse Accelerometer and Gyroscope

- State vector

$$
\begin{bmatrix} x \\ y \\ \varphi \end{bmatrix}
\begin{array}{l} \text{x-position} \\ \text{y-position} \\ \text{angle} \end{array}
$$




- System model

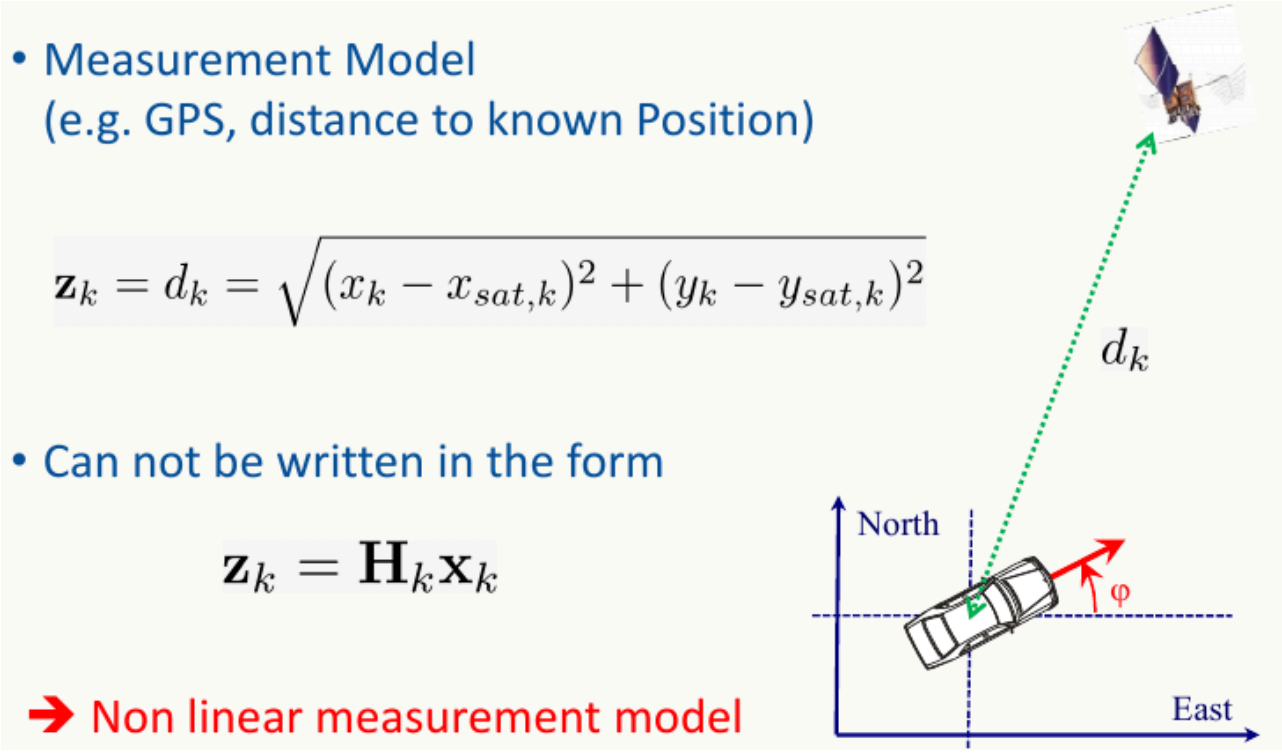
travelled distance (e.g. from odometer)

$$
\begin{bmatrix} x_k \\ y_k \\ \varphi_k \end{bmatrix} = \begin{bmatrix} x_{k-1} \\ y_{k-1} \\ \varphi_{k-1} \end{bmatrix} + \begin{bmatrix} \cos\varphi_{k-1} \cdot \boxed{\Delta t \cdot v_k} \\ \sin\varphi_{k-1} \cdot \Delta t \cdot v_k \\ \boxed{\Delta t \cdot \omega_k} \end{bmatrix}
$$

relative angle (e.g. from gyroscope)

- Information about relative motion can be used as control input in the system model
- Can not be written in the form: $\boldsymbol{x}_k = \boldsymbol{\Phi}_k \boldsymbol{x}_{k-1} + \boldsymbol{B}_k \boldsymbol{u}_k$

- -> **Non-linear system model!**




## Non Linear Filtering

- Predicted state is function of last state, control input and noise
  - $\boldsymbol{x}_k = f(\boldsymbol{x}_{k-1}, \boldsymbol{u}_k, \boldsymbol{w}_k)$
- Measurement is a function of the state and noise
  - $\boldsymbol{z}_k = h(\boldsymbol{x}_k, \boldsymbol{v}_k)$

**Prediction**

$$\begin{aligned} \mathbf{x}_k^- &= f(\mathbf{x}_{k-1}, \mathbf{u}_k) \\ \mathbf{P}_k^- &= \mathbf{\Phi}_k \mathbf{P}_{k-1} \mathbf{\Phi}_k^T + \mathbf{Q}_k \end{aligned}$$

**Correction**

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}_k^T (\mathbf{H} \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k (\mathbf{z}_k - h(\mathbf{x}_k^-)) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- \end{aligned}$$

$$\mathbf{\Phi}_k = \frac{\partial f(\mathbf{x}, \mathbf{u}, \mathbf{w})}{\partial \mathbf{x}} \Big|_{\mathbf{x} = \hat{\mathbf{x}}_{k-1}, \mathbf{u} = \mathbf{u}_k} \qquad \mathbf{H}_k = \frac{\partial h(\mathbf{x}, \mathbf{v})}{\partial \mathbf{x}} \Big|_{\mathbf{x} = \hat{\mathbf{x}}_k^-}$$

Example 1 (**measuring** path increment, angular rate and gps , **modeling** constant angular rate and constant velocity)

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \\ \dot{\varphi}_k \\ v_k \end{bmatrix} = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1} \Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1} \Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1} \Delta t \\ \dot{\varphi}_{k-1} + w_{\dot{\varphi}} \Delta t \\ v_{k-1} + w_v \Delta t \end{bmatrix}$$

$$\begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ \Delta s_k \\ \omega_k \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}_k$$

system noise

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{\dot{\varphi}}^2 \Delta t^2 & 0 \\ 0 & 0 & 0 & 0 & \sigma_v^2 \Delta t^2 \end{bmatrix}$$

Example 2 (**measuring** acceleration, angular rate, and gps, **modelling** constant angular rate and constant acceleration)

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \\ \dot{\varphi}_k \\ v_k \\ a_k \end{bmatrix} = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1} \Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1} \Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1} \Delta t \\ \dot{\varphi}_{k-1} + w_{\dot{\varphi}} \Delta t \\ v_{k-1} + a_{k-1} \Delta t \\ a_{k-1} + w_a \Delta t \end{bmatrix}$$

$$\begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ a_k \\ \omega_k \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \mathbf{x}_k$$

system noise

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{\dot{\varphi}}^2 \Delta t^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & & 0 \sigma_a^2 \Delta t^2 \end{bmatrix}$$

Example 3 (**measuring** path increment, angular rate, and gps, using ang. rate and path inc. as **control input**)

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \end{bmatrix} = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1})(\Delta s_k + w_{odo}) \\ y_{k-1} + \sin(\varphi_{k-1})(\Delta s_k + w_{odo}) \\ \varphi_{k-1} + (\omega_{gyro,k} + w_{gyro}) \Delta t \end{bmatrix}$$

$$\begin{bmatrix} x_{gps,k} \\ y_{gps,k} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x}_k$$

system noise (from contol input)

$$\mathbf{Q} = \begin{bmatrix} \sin^2(\varphi_{k-1}) \sigma_{odo}^2 & \cos(\varphi_{k-1}) \sin(\varphi_{k-1}) \sigma_{odo}^2 & 0 \\ \cos(\varphi_{k-1}) \sin(\varphi_{k-1}) \sigma_{odo}^2 & \cos^2(\varphi_{k-1}) \sigma_{odo}^2 & 0 \\ 0 & 0 & \sigma_{gyro}^2 \Delta t^2 \end{bmatrix}$$

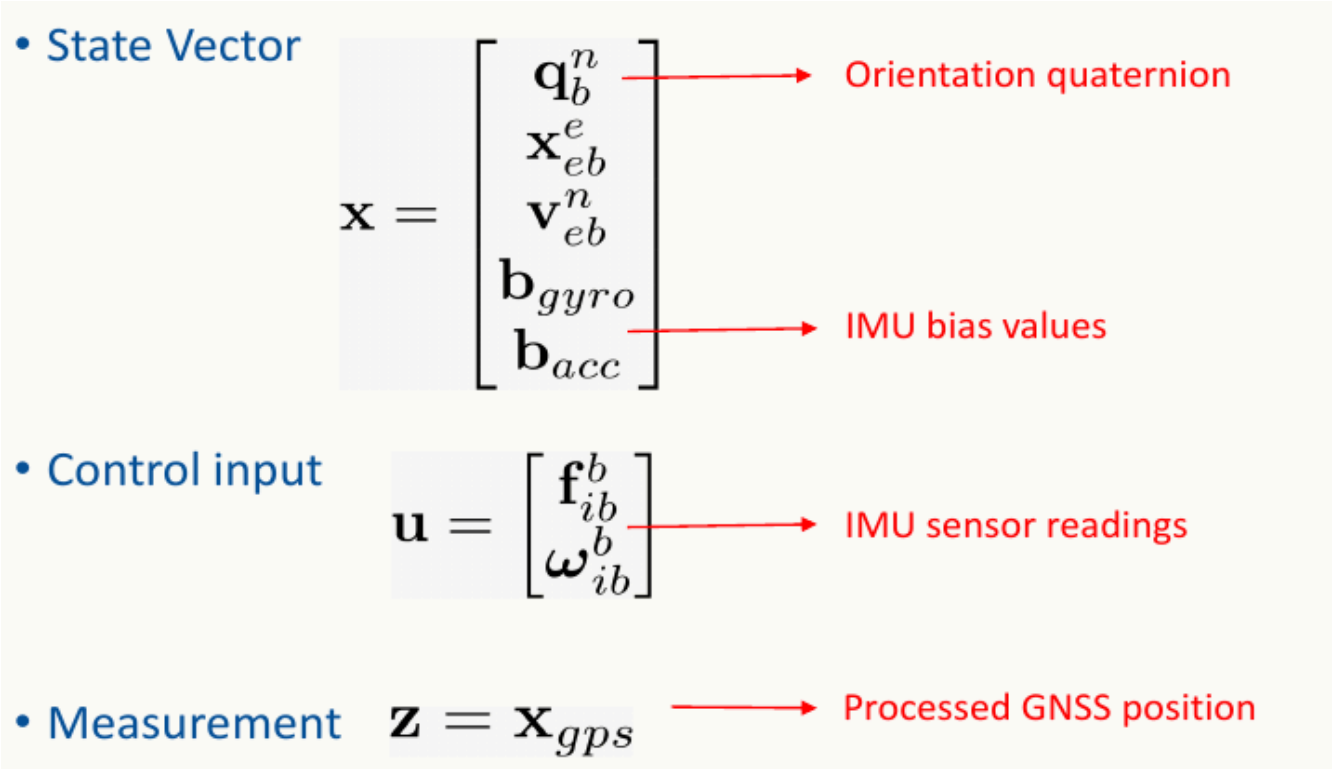
## Inertial Navigation Systems

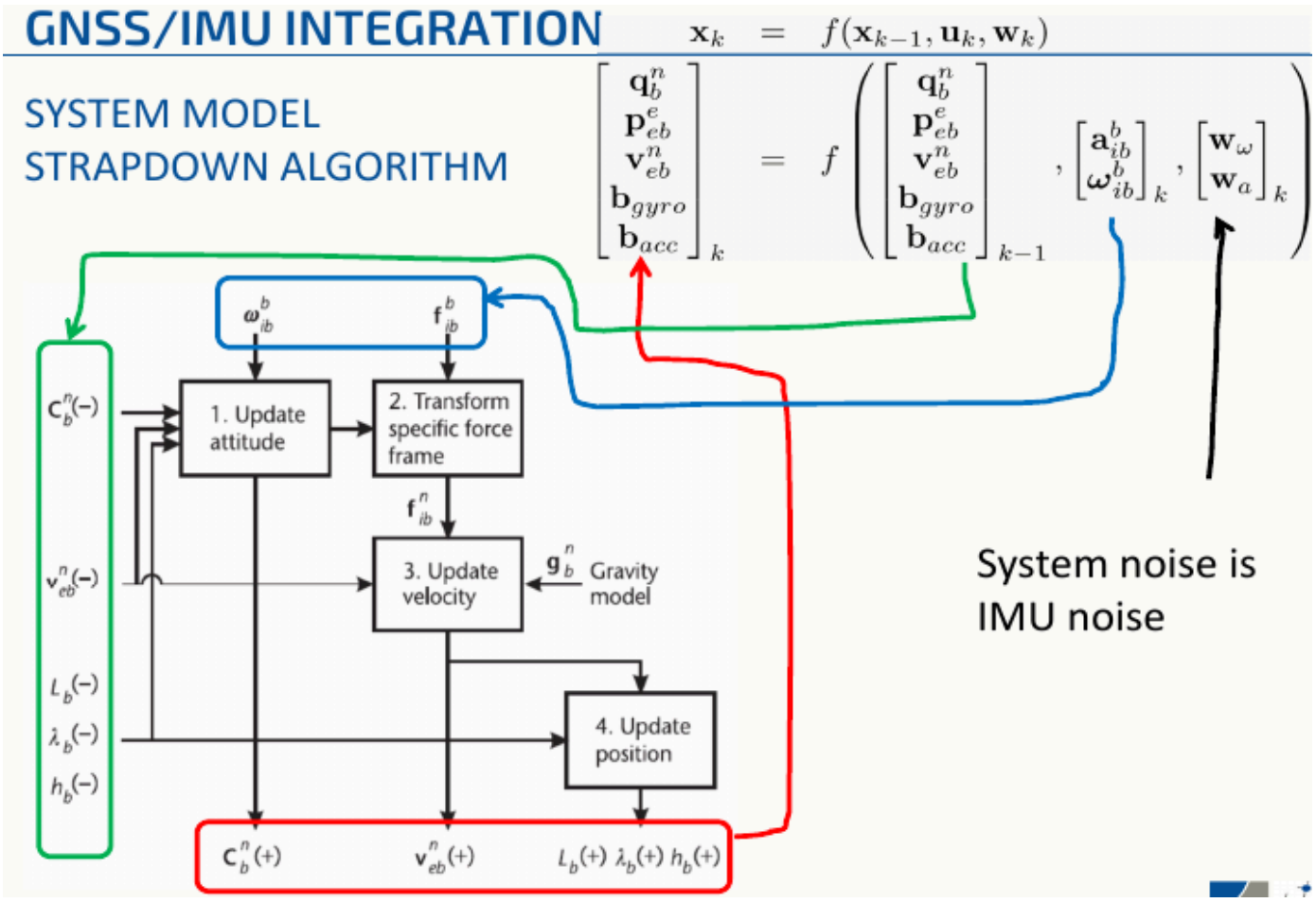
- 3 axis gyro
- 3 axis accelerometer
- GNSS receiver with 1 or 2 antennas
- **Goal**:
  - 3D position
  - 3D Orientation
  - 3D Velocity

## GNSS/IMU Integration

- Estimate position & orientation of the system in all dimensions
- Use inertial sensors (acc + gyro) as control input:
  - Strapdown inertial integration as system model
- Error of acc + gyro: Time **dependent systematic errors** (bias)
  - Estimate biases as part of the state vector
- Use GNSS in measurement model

- State Vector

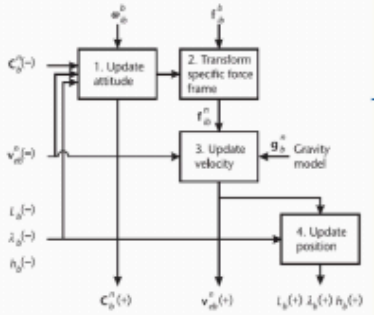


## GPS/IMU INTEGRATION

### MEASUREMENT MODEL

$$\mathbf{x} = \begin{bmatrix} \mathbf{q}_b^n \\ \mathbf{x}_{eb}^e \\ \mathbf{v}_{eb}^n \\ \mathbf{b}_{gyro} \\ \mathbf{b}_{acc} \end{bmatrix}$$




$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \boxed{\mathbf{R}})^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - \boxed{h(\mathbf{x}_k^-)}) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \end{aligned}$$

- Measurement is only GPS Position $x_{gps}, y_{gps}, z_{gps}$

$$\mathbf{z}_k = \begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ z_{gps,k} \end{bmatrix} = h(\mathbf{x}_k) = \mathbf{H}\mathbf{x}_k = \begin{bmatrix} \dots & 1 & 0 & 0 & \dots \\ \dots & 0 & 1 & 0 & \dots \\ \dots & 0 & 0 & 1 & \dots \end{bmatrix} \mathbf{x}_k$$

- Measurement Noise

$$\mathbf{R} = \begin{bmatrix} \sigma_{gps,x}^2 & 0 & 0 \\ 0 & \sigma_{gps,y}^2 & 0 \\ 0 & 0 & \sigma_{gps,z}^2 \end{bmatrix}$$

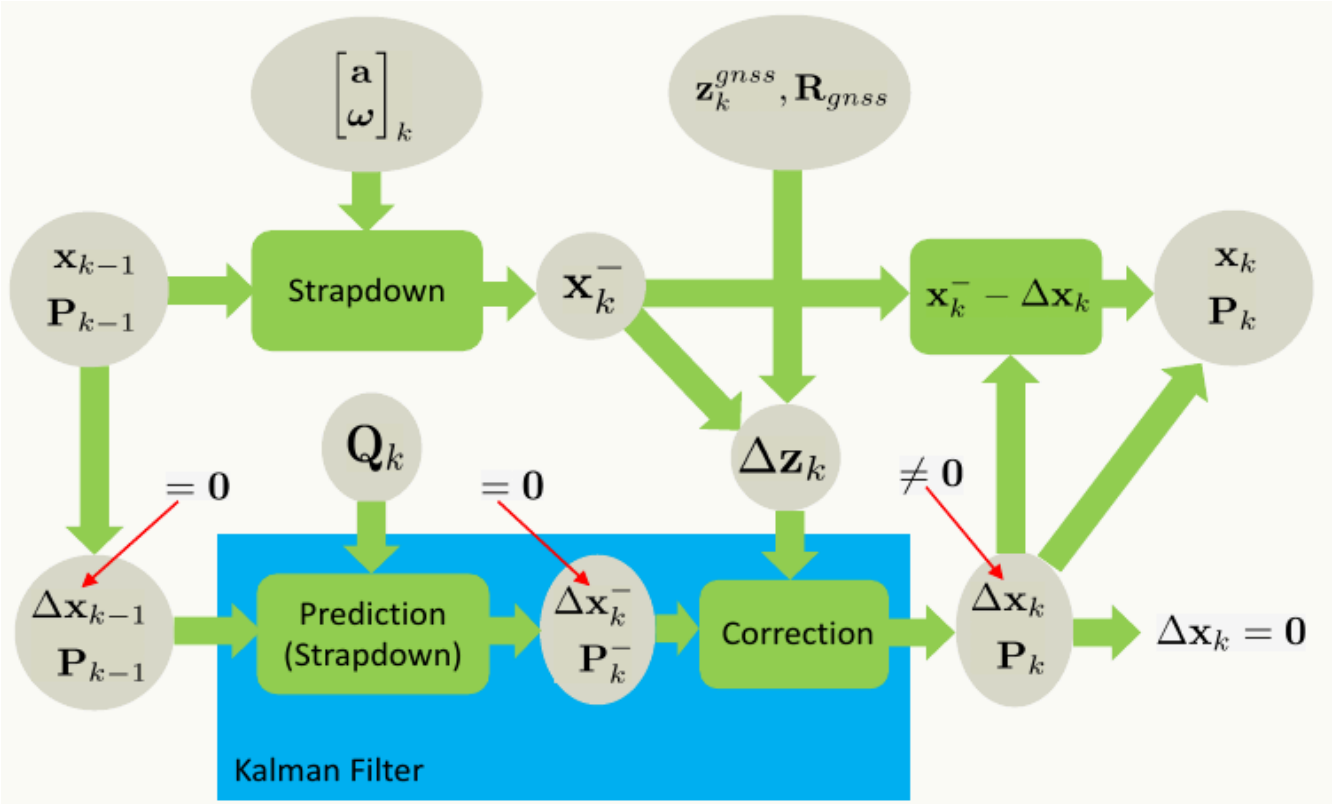
- But Remember: GPS noise is not white
  - R is wrong

### Tightly Coupled Vs. Loosely Coupled

- Using a **processed GPS position** in the measurement model:
  - **Loosely Couple Integration**
- Using **raw** GNSS observation (carrier phases, pseudo ranges):
  - **Tightly Coupled Integration**
  - GNSS processing (+ ambiguity fixing) within filter
  - Better ambiguity fixing, because IMU data Integration
  - Included Cycle slip detection
  - Observation from a small number of satellite (<4) can be integrated into filter
  - More complex filter algorithm
- Comparable results for high satellite visibility

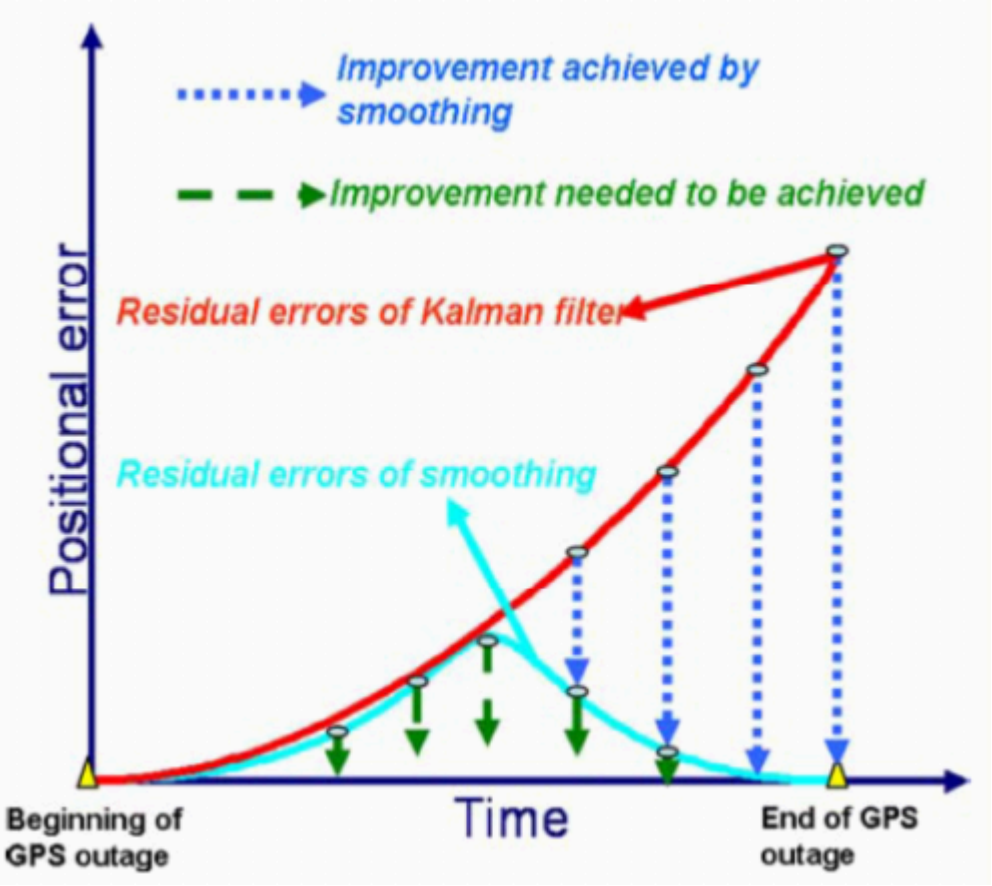
## Error State Filtering

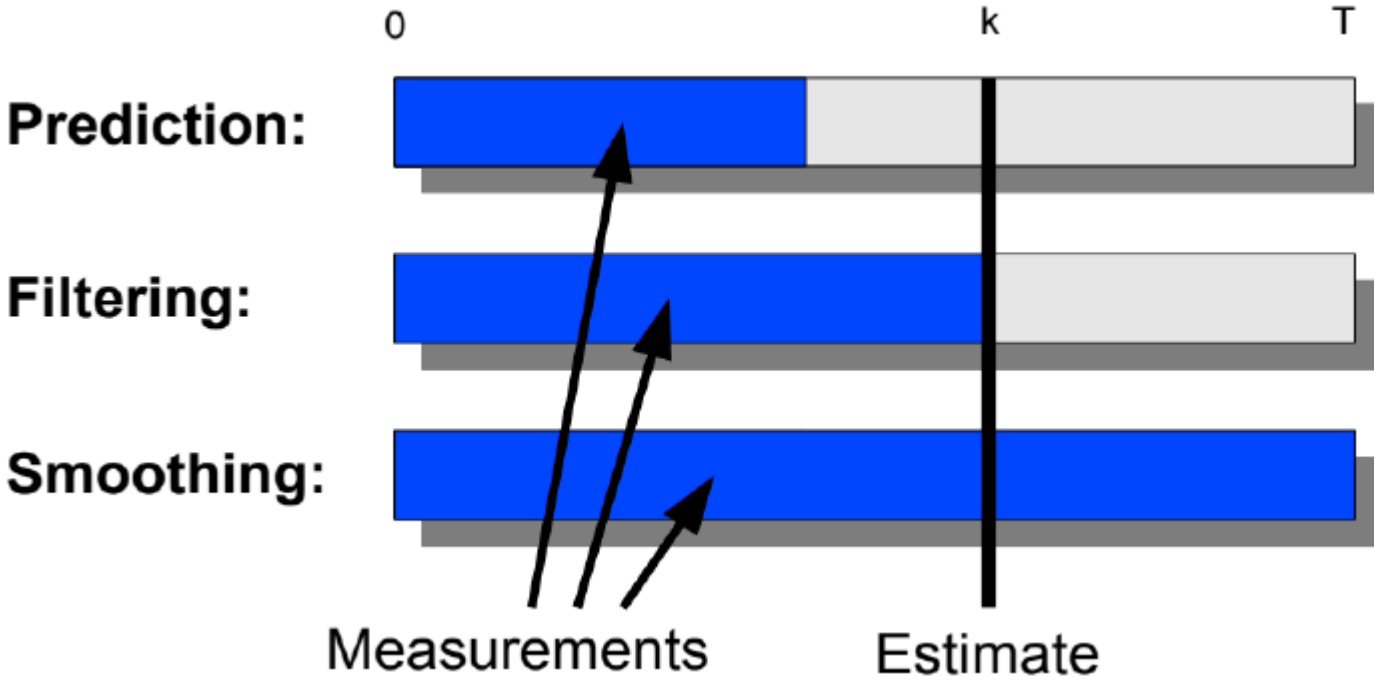
- GNSS/IMU Fusion Error State KF (ESKF)




- Error State Filtering
  - Theoretically identical
  - Potentially more numerically stable
  - Common realisation of GNSS/IMU Fusion Kalman Filter
  - Error dynamics are lower than motion dynamics
    - Linearisation is not that critical
    - Error Filter update rate can be smaller

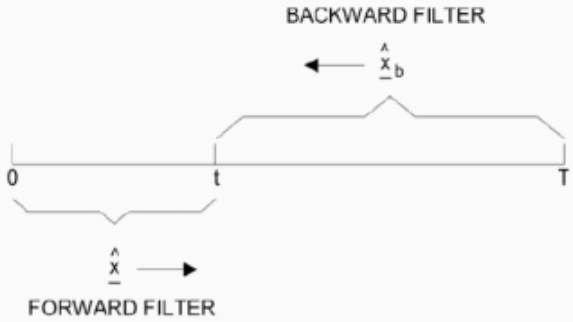
## Kalman Smoothing




**Forward-Backward Filter**

## FORWARD-BACKWARD FILTER

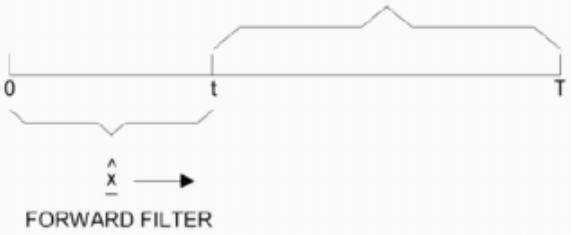



Weighted mean of forward and backward filter

$$
\begin{aligned}
\mathbf{x}_{S,k} &= K_{F,k}\mathbf{x}_{F,k} + K_{B,k}\mathbf{x}^{-}_{B,k} \\
&= K_{F,k}\mathbf{x}_{F,k} + (1 - K_{F,k})\mathbf{x}^{-}_{B,k} \\
K_{F,k} &= P^{-}_{R,k}(P_{F,k} + P^{-}_{B,k})^{-1} \\
P_{S,k} &= \left[(P_{F,k})^{-1} + (P^{-}_{B,k})^{-1}\right]^{-1}
\end{aligned}
$$

### RTS Smoother

1) Start with a forward filter from 0 to T
2) Run a „smoother“ from T to 0

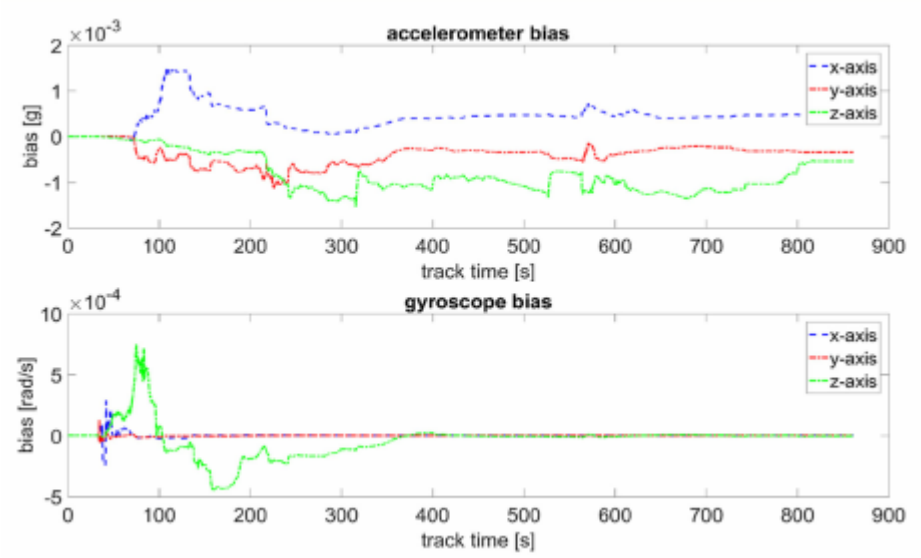


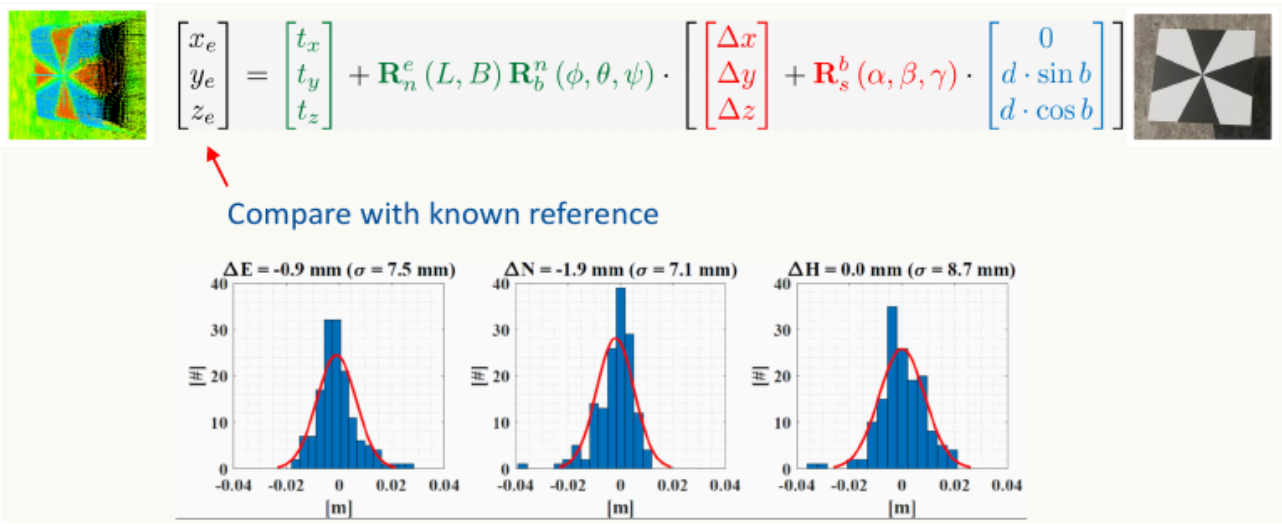

$$
\begin{aligned}
\mathbf{x}_{S,k} &= \mathbf{x}_{F,k} + A_k(\mathbf{x}_{S,k+1} - \mathbf{x}^{-}_{F,k+1}) \\
A_k &= P_{F,k}\Phi_k^T\left[P^{-}_{F,k+1}\right]^{-1} \\
P_{S,k} &= P_{F,k} + A_k(P_{S,k+1} - P^{-}_{F,k+1})
\end{aligned}
$$

More efficient. Only need to store forward states and covariances

## Trajectory Estimation / Quality Evaluation

- Look at **Covariance** of the Estimate $P_k$
  - Does **not depend on measurements**
  - Only valid, if all models and uncertainties are **correct**
  - **No systematic errors** included

- Look at **Innovation** $(z_k - H\boldsymbol{x}_k^-)$
  - Gives idea on consistency between models and observations
  - Not clear if inconsistencies come from **wrong measurement** or **predictions**
- Look at Estimated States
  - Nice looking position result can also appear if model completely wrong and measurement are good
  - E.g. Estimated bias values show consistency of models
  - 

- Compare with reference Solution
  - Needs synchronisation between reference and estimation results
  - Difficult to realise in outdoor scenarios (especially orientation)
  - Error depends on motion, environment, time -> impossible parametrisation
- Indirect Evaluation
  - Includes many other errors
  - Assignment of error to specific sources is difficult
  - 

# Evaluation without Reference Information

- **Track comparison**: RTS smoother without and with GNSS outages
- **GNSS Quality**
  - Dependent on Number of GNSS satellites
- Difference to GNSS data
- Bias estimation